



AIRBUS

A300-600R / A310

SOF



STANDARD OPERATING PROCEDURES

A 300-600R A310

DOCUMENTATION CONTROL PAGE

**Document Title: AIRBUS 300/600R / A310 STANDARD OPERATING
PROCEDURES**

Effective Date: As signed

Revision No. : 04

	A300-600R / A310 FLEET DIRECTOR
Prepared By	Capt. FAKHRZADEH Signature: <u>Fakhrzadeh</u>
Checked and Reviewed By	Flight Safety and Operations Standard Capt. Signature: <u>Sadegh Roshan</u>
Accepted By	Deputy, Managing Director, Operations Capt. AKBARI Signature: <u>AKBARI</u>

PAGE No	REASONS OF CHANGE
06-09	EXTERIOR INSPECTION ASTRIKS REVOED
18	PF must insert wind information in FMS
26	BRAKE CHECK procedure changed
30	"CABIN CREW T/O POSITION" will be done by PM
34	CRUISE BERIFING
38	"CABIN CREW PREPARE FOR APPROACH" will be done by PF
40	BARO SETTING/"CABIN CREW LANDING POSITION" will be done by PM

Revisions to this Manual shall be made by all authorized users without delay. After inserting the revision enter the appropriate date in the revision sheet below. The compliance has to be signed under "ENTERED BY".

REVISION NUMBER	DATE ENTERED	ENTERED BY
Version 01	August 2013	First issue
Revision 02	October 2013	Fleet Office
Revision 03	May 2015	Fleet Office
Revision 04	December 2018	Fleet Office

[illegible]

SAFETY EXTERIOR INSPECTION

- Items marked by (*) are the only steps to be completed during a transit stop.
- Completion ensures that aircraft and surroundings are not obviously unsafe for operations.
- On arriving at the aircraft check for obstructions in the vicinity, engineering activity, refueling etc...

PF

* ALL WHEEL CHOCKSCK IN PLACE
* L/G DOORSCK POSITION
* APU AREA.....CK

PM

* ALL WHEEL CHOCKS.....CK IN PLACE
* L/G DOORS.....CK POSITION
* APU AREA.....CK

- Observe APU inlet and outlet are clear.
- **CAUTION:** Do not pressurize the green hydraulic system without ground clearance if any gear door is open.

PRELIMINARY COCKPIT PREPARATION

PF

FLIGHT DECK SECURITY CHECKCOMPLETE

PM

- Refer to flight deck security checklist.
- The following procedure, performed by the PF, ensures that required safety checks are made prior to the application of electrical power to avoid inadvertent systems operation and danger to aircraft and personnel. Included is APU starting and establishment of electrical and pneumatic power.

IGN CK OFF
WIPERS CK OFF
L/G LEVER.....CK DN
THROTTLE LEVERS.....CK IDLE
REV LEVERSCK STOWED
FUEL HP VLVS.....CK OFF
ATC XPDR..... CK STBY
WX RADAR.....CK OFF / TEST
C/B PANELS CK

- Check all C/B's are set. If a C/B is found tripped, coordinate with maintenance for reset.
- **BATTERIES and EMERGENCY INVERTER:**
- BAT 1,2 and 3 Check OFF
- BAT 1,2 and 3 voltage Check Above 25 V
 - a) If one battery is below 25 V connect external power before attempting to start the APU.

BAT / EMER INV..... AUTO / CK

- a) Check FLOW BARS in line and observe DOME light is illuminated.
- b) AC EMER ON INV light Check ON
- c) DC ESS ON BAT light Check ON
- d) AC VOL/FRQ rotary sel EMER
- e) Check INV parameters in white range.
- **CAUTION:** In case of emergency inverter malfunction, switch off batteries before any other action to avoid damage to the electronic equipment.
- **HYDRAULIC:**
- **CAUTION:** Do not pressurize the green hydraulic system without ground clearance.

PRELIMINARY COCKPIT PREPARATION	
PF	PM
COCKPIT EMERGENCY LIGHTING CK	
• APU FIRE PROTECTION:	
APU FIRE CK / TEST	
• APU FIRE handle Check IN and Latched • LOOP A and B pb Check IN • SQUIB TEST pb Press • AGENT SQUIB It Check Illuminated • LOOP TEST pb Press and Hold a) Check: LOOP A It comes on with LOOP warning activation. After few seconds, LOOP B It also illuminates associated with APU FIRE warning activation and APU FIRE handle illumination. b) LOOP TEST pb Release c) Check LOOP A warning and APU FIRE warning activation disappear. After few seconds LOOP B warning disappears. • ELEC: a) If AVAIL Lt illuminated:	
EXT PWR ON	
b) If AVAIL Lt extinguished. • APU START: a) If EXT PWR ON It extinguished : b) APU MASTER SW ON c) APU START ON d) ON It illuminates. Blue ACCEL light illuminates and indicates that APU is accelerating. e) At 50 % RPM : START PB switch is released automatically, ON white light extinguishes. f) At 95% RPM : AVAIL It illuminates. g) APU GEN on line : ECAM APU page appears after 10 seconds. If required, adjust brightness on ECAM control panel.	
PARK BRK / PRESS ON / CK	
• Check ACCU PRESS and recharge if necessary. The ACCU PRESS indication must be in the green band. • If required, use the electric pump on yellow hydraulic system to recharge the brakes accumulators. • Check PARKING BRAKE pressure is 1500 PSI or more. • PARKING BRAKE must be applied before performing the Exterior Inspection to allow removal of chocks and to check BRAKE wear indicators.	

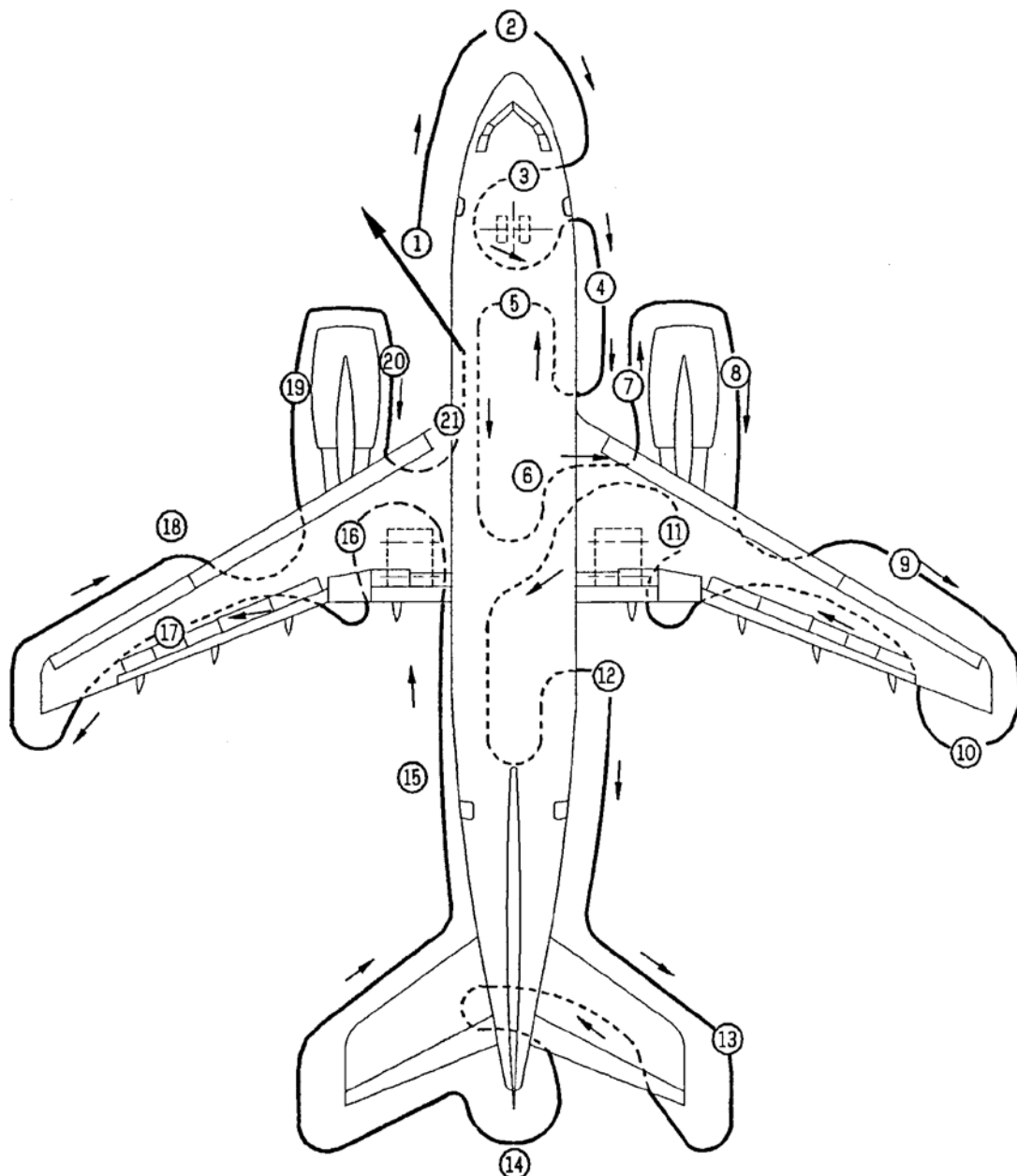
PRELIMINARY COCKPIT PREPARATION

PF	PM
EXT LTSSET COCKPITLTS.....ASRQRD	* EXT WALK AROUND.....PERFORM
Set STBY COMPASS, DOME, STORM and PANEL lights as required. DOME Light should be selected ON, Since it is the only lighting source in EMER ELEC configuration. DIM position is recommended for takeoff.	
ELEC PANEL CK	
Scan ELEC PWR panel : no amber light illuminated except GEN 1 and GEN 2 FAULT lights.	
LEFT INNER PUMP 2.....NORM	
If there is sufficient fuel in the left inner tank, this avoids fuel depletion from the outer tanks which would lead to increased refueling time.	
PROBE / WINDOW HEATCK OFF VENT PANEL CK	
- Check all lights OFF. NOTE: On ground OVBD green flow bar in line.	
ANN LT TEST	
- Check that all cockpit annunciator lights come on bright. - The windshear warning is triggered. WINDSHEAR appears in red on the PFD for approximately 15 seconds, (If PFD is on). - Audio warning "WINDSHEAR" is announced three times. - WINDSHEAR ALERT NOT AVAILABLE message is displayed on ECAM STATUS page.	
APU BLEED ON	
- Do not use APU bleed if a ground personnel confirms that ground air is connected. Pilots should also check the ECAM BLEED page to determine whether an HP ground air is connected (pressure in the system). - PACK VALVES 1 and 2 pb switchesCheck ON, Check Flow bars in line	
SPD BRKCK RET SLAT/FLAP.....CK POSITION	
CAUTION: If flight control surface positions do not agree with the control handle positions, check with the maintenance personnel before applying hydraulic power. - Check SPD BRK in RETRACT position and DISARM. - If SLAT/FLAP handle position disagrees with slats/flaps position, set handle to corresponding position. - At this stage, with no hydraulic pressure and with the slats/flaps handle in 0/0 position, the KRUGER NOT RETRACTED warning on the ECAM and the KRUGER light on the SFPI may be triggered. NOTE: The notch flaps may droop from the stowed position as a result of low hydraulic pressure. This does not require maintenance action.	

PRELIMINARY COCKPIT PREPARATION	
PF	PM
LATERAL PANEL:	
MTP CK	
- Check ANN LT sel in OFF position. - Check no light illuminated on lateral panel.	
1 st OBS ACPPA	
- Select P.A. reception on audio selector of lateral panel, with volume selection in approx 12 o'clock position. - This selection will allow recording of cabin attendants announcements on the cockpit voice recorder.	
EMER EQPTCK	
Check: - Life jackets stowed. - Asbestos gloves stowed. - Axe stowed. - Flashlights stowed. - Smoke goggles stowed. - Portable fire extinguisher lock wired and pressure within the green area. - Oxygen masks stowed. - Portable oxygen equipment and full face masks, bottle pressure within the green area and adapters stowed underneath right aft window. - RH and LH evacuation handles stowed. - L/G gravity extension handle stowed. - Smoke hoods.	
GEAR PINS CK	
Check three gear pins on board and stowed.	

EXTERIOR WALK AROUND

- **EXTERIOR WALK AROUND:**



EXTERIOR WALK AROUND

GENERAL

The exterior inspection ensures that the general condition of the aircraft is satisfactory and that the visible aircraft components are safe for flight:

- There is no impact damage to the structure
- There is no evident fuel, oil, or hydraulic leak
- All ground access doors are closed
- The flight crew must perform a complete walkaround before each flight
- The parking brake must be set ON during the exterior walkaround, in order to enable the flight crew to check brake wear indicators.
- **CAUTION:** IF A L/G DOOR IS OPEN CONTACT MAINTENANCE CREW BEFORE APPLYING HYD POWER.

1) LEFT FORWARD FUSELAGE

- Static portsCLEAR
- Exterior lightsCONDITION
- Angle of attack probeCONDITION
- Toilet service access doorCLOSED

2) NOSE SECTION

- Pitot probes and coversCHECK/REMOVED
- TAT probes.....CONDITION
- Radome and latchesCHECK
- Oxygen compt venturiCHECK
- Oxygen over pressure diskGREEN

3) NOSE GEAR

- Nose wheel chocksIN PLACE
- Wheels and tiresCONDITION
- Nose gear structureCONDITION
- Taxi lightsCONDITION
 - External electric PWR access door, (If EXT PWR not necessary)CLOSED
- Nose wheel steering selector and by-pass pinCHECK
- Proximity detectorsCLEAN
- Hydraulic lines and control cablesCONDITION
- Wheel wellCHECK
- Safety pinREMOVED

4) RIGHT FORWARD FUSELAGE

- Cargo doorsCHECK
- Cargo loading control doorCHECK
- Cargo door selector access doorCHECK
- Exterior ltsCONDITION
- Static portsCLEAR
- Angle of attack probesCONDITION
- Lower fuselage access doorsCLOSED

EXTERIOR WALK AROUND

5) LOWER CENTER FUSELAGE

- Extract valve / fan**OPEN AND RUNNING**
- Antennas**CONDITION**
- Ground air conditioning door**CLOSED**
- Fresh water service door**CLOSED**
- Ram air inlet flap**CLOSED**
 - Ground air conditioning / engine starting door**CLOSED**
- Pack air intakes**CLEAR**
 - Landing gear door opening on ground access doors**CLOSED**
- * - Refueling electric control panel access door.....**CLOSED**
- Packs air outlets and flaps **CLEAR**

6) RIGHT CENTER WING

- Wing ventilation air intake**CLEAR**
- RH Inner fuel tank water drain valves**CHECK**
- Drip sticks **SECURED**
- Landing light**CONDITION**
- Yellow hydraulic access door**CLOSED**
- Ram air turbine doors**CLOSED**
- Slat 1 and Kruger**CONDITION**

7) ENGINE 2 LEFT SIDE

- Cowlings**CLOSED/LATCHED**
- Drain mast**CHECK**
- Hydraulic filter access door**CLOSED**
 - Starter valve manual override access door**CLOSED**
- Engine inlet and fan blades**CHECK**

8) ENGINE 2 RIGHT SIDE

- Anti-ice exhaust**CLEAR**
- Engine oil access door**CLOSED**
- Turbine exhaust**CLEAR**
- Pylon overpressure door**CLOSED**
- Fire agent bottles access door**CLOSED**
- Pylon drains**CHECK**

9) RIGHT WING LEADING EDGE

- Slats 2 and 3**CONDITION**
- Refuel plugs**CHECK**
- Drip sticks**SECURED**
- RH outer fuel tank water drain valves**CHECK**
- Surge tank air inlet**CLEAR**
- Fuel Vent. 0/press disc**INTACT**
- Wing tip**CHECK**
- Navigation and Strobe Lights **CHECK**

EXTERIOR WALK AROUND

10) RIGHT WING TRAILING EDGE

- Static dischargersCHECK
- * - Control surfacesCHECK
- Flap track fairing'sCHECK
- Flaps.....CONDITION

11) MAIN RIGHT LANDING GEAR AND FUSELAGE

- Wheel chocksREMOVED
- Wheels and tiresCONDITION
 - Brakes and brake wear indicatorsCONDITION
- Proximity sensorsCLEAN
- Hydraulic linesCHECK
 - Pitch damper pressure drop indicatorCHECK
- Landing gear structureCHECK
- Gear uplock hookCONDITION
- Down lock springCHECK
 - Landing gear down lock safety pin REMOVED
- Wheel well CHECK
- Landing gear doorCLOSED
- Fuselage Fuel drain mastCHECK
- Water drain mastCHECK
- Antennas and beaconCHECK

12) RIGHT AFT FUSELAGE

- Cargo loading control access doorCHECK
 - Cargo compartment door hydraulic selector access doorCHECK
 - Cargo door lock-unlock control leverCHECK
- Cargo doorsCHECK
- Outflow valveCHECK
- Toilet service access doorCHECK
- Tail cone access doorCLOSED
 - RH landing gear down lock indicator pin EXTENDED
- Tail skidCHECK

13) TAIL SECTION

- Stabilizer and elevatorCHECK
- Fin and rudderCHECK
- Static dischargersCHECK
- Fuel vent O/press discINTACT
- Trim tank water drain valvesCHECK
- Tail vent tank water drain valveCHECK
- Drip stickSECURED

14) APU

- APU doorsCLOSED
- APU air intakeOPEN
- APU drain and ventsCHECK
- Oil cooler exhaustCHECK
- APU exhaustCLEAR
- APU fire agent discharge overpressured disc CHECK PRESENT

15) LEFT AFT FUSELAGE

- LH landing gear downlock indicator pinEXTENDED
- Stabilizer and elevatorCHECK
- Fresh water drainCHECK
- Green hydraulic access doorCLOSED

EXTERIOR WALK AROUND

16) MAIN LEFT LANDING GEAR

- Wheel well CHECK
- Landing gear door CLOSED
- Wheels chocks REMOVED
- Wheels and tires CONDITION
- Brakes and brake wear indicator CONDITION
- Proximity sensors CLEAN
- Hydraulic lines CHECK
- Pitch damper pressure drop indicator CHECK
- Landing gear structure CHECK
- Gear uplock hook CONDITION
- Downlockspring CHECK
- Landing gear downlock safety pin REMOVED

17) LEFT WING TRAILING EDGE

- Flaps CONDITION
- Flap track fairings CHECK
- Control surfaces CHECK
- Static dischargers CHECK

18) LEFT WING LEADING EDGE

- Navigation and strobe lights CHECK
- Wing tip CHECK
- Surge tank air inlet CLEAR
- Fuel vent O/press disc INTACT
- LH outer fuel tank water drain valves CHECK
- Drip sticks SECURED
- Refuel plugs CHECK
- Slats 2 and 3 CONDITION

19) ENGINE 1 LEFT SIDE

- Pylon drains CHECK
- Fire agent bottle access door CLOSED
- Pylon overpressure door CLOSED
- Cowlings CLOSED/LATCHED
- Drain mast CHECK
- Hydraulic filter access door CLOSED
- Starter valve manual override access door CLOSED
- Engine inlet and fan blade CHECK

20) ENGINE 1 RIGHT SIDE

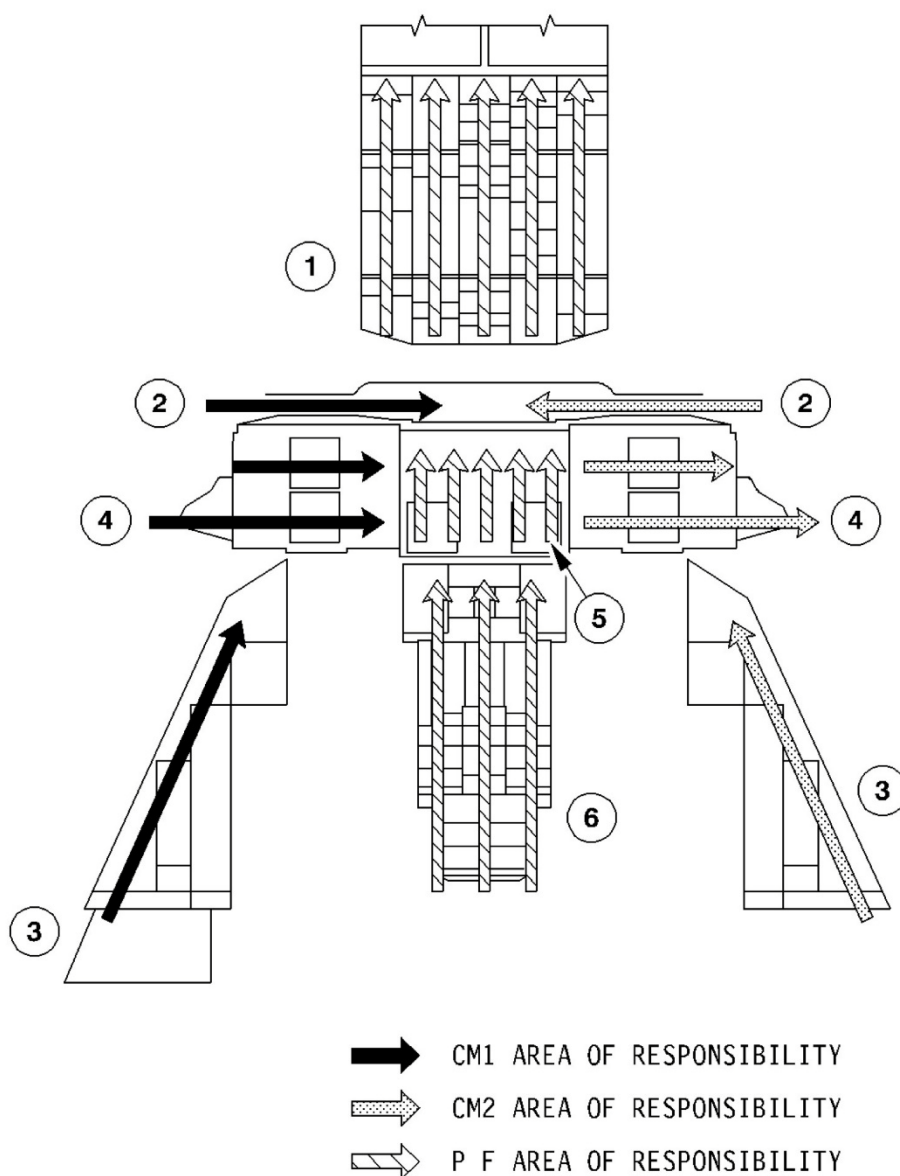
- Anti ice exhaust CLEAR
- Engine oil access door CLOSED
- Turbine exhaust CLEAR

21) LEFT CENTER WING

- Slat 1 and Kruger CONDITION
- Blue hydraulic access door CLOSED
- Landing light CONDITION
- Drip sticks SECURED
- LH inner fuel tank water drain valves CHECK
- Wing ventilation air intake CLEAR

COCKPIT PREPARATION

- INTRODUCTION:
- Items marked by (*) are the only steps to be completed during a transit stop.
- The cockpit preparation should be performed by the **PF** and **PM** according to panel scan sequence, defined below, and the task sharing defined in the ORH.
- DOCUMENTATION AND MAINTENANCE
- On entering the aircraft, obtain the technical (maintenance) log, verify the certificate of maintenance and daily inspection (or similar) are up to date and signed. Check the deferred or carried forward defects. If the refueling has already been performed check the uplift.
- PANELS SCAN SEQUENCE:



COCKPIT PREPARATION

PF

PM

- **NOTE1:** it is a general rule to extinguish all white lights for all systems during the scan sequence, these actions are therefore not listed here.
- **NOTE2:** fuel pumps may be left off until refueling is completed if required.
- **NOTE3 :** when the fuel pumps are set to the norm position closely monitor the Lo PR and fault lights and Make sure all the lights turn off in usual timeframe

DATABASE VALIDITY CK
* **FMS LAT / LONG** CK / CORRECT
* **IRS** NAV

- Press REF key and display AIRCRAFT STATUS page.
- Check DATABASE validity and stored WPT/NAVAID if any, review stored data for deletion decision.
- Check PERF FACTOR agrees with Flight plan.
- The IRS MSUs will normally be OFF when entering the cockpit, selecting each MSU to NAV will start the IRS alignment process.
- If the MSUs are still in NAV mode, select to OFF for more than 10 seconds, then back to NAV for a complete realignment, which lasts approx 10 minutes.
- ALIGN MODE will be illuminated.
- **IRS ALIGNMENT:**
- Perform a complete alignment if:
 - a) It is the first flight of the day.
 - b) The GPS is not available, and long segments poor radio NAVAID coverage airspace are expected.
- For other flights, perform a fast alignment, if the residual ground speed is greater than 5 knots. The alignment is not necessary, if the residual ground speed is less than 5 knots.
- **NOTE :**
 - a) The IRS outputs are used by many systems of the aircraft, so it is essential to align the IRS as early as possible to provide data to the related systems.
 - b) Check IRS ALIGN MEMO on ECAM is no longer displayed before moving the aircraft.
 - c) Do not initiate a rapid realignment within less than 3 minutes after normal alignment.
 - d) If an IRS alignment discrepancy occurs (ALIGN light flashes on MSU), check present position is correct and re-enter the present position using the same tuning source (FMS CDU or ISDU) as the one previously used.

* **ALIGN IRS PROMPT** PRESS

- When any of the IRUs are in the align mode the prompt ALIGN IRS will be displayed. Press the ALIGN IRS prompt to align the IRS on the LAT/LONG entered in the CDU, as soon as MSU mode selectors are selected to NAV.
- If another page is selected other than INIT A, the message ALIGN IRS remains in the scratchpad.
- **NOTE:** If IRS had previously been aligned on ground at the same LAT/LONG, then the ALIGN IRS prompt will not be displayed. If the LAT/LONG are changed, the ALIGN IRS prompt is displayed.
- ISDU Check valid co-ordinates have been sent.

* **SIGNS** SET

- NO SMOKING ON
- SEAT BELTS FFO

COCKPIT PREPARATION

PF	PM
* CALLS.....CK	
Check CALL Its extinguished. Reset as required.	
HYD QTYS.....NORM RANGE	
- Check the 3 fluid level indications are normal and within range. NOTE: On the ground, system not pressurized, quantity indication Needle should be in the outer green sector for any OAT. NOTE: For OAT below -10°C (above 20°C), the needle may be below (above) the upper small green arc down (up) to two needles width.	
* FLT RCDR GND CTL.....ON	
Check DFDR and DFDAU Lights extinguished.	
ISDU OFF	
Check SYS rotary selector set to OFF.	
* EXTLTS.....AS RQRD	
Check STROBE in AUTO position and BEACON in OFF position.	
* PITCH TRIM / YAW DAMPER / ATS..... ON	
- PITCH TRIM 1 + 2 ON - YAW DAMPER 1 + 2 ON - ATS 1 (+ 2 If installed) ON	
ENG 1 FIRE PROTECTION.....CK / TEST	
- ENG 1 FIRE handle.....Check IN and latched - SQUIB TEST pb Press - AGENT SQUIB 1 + 2 Its Check Illuminated - LOOP TEST pb Press and Hold a) LOOP A It comes on associated with LOOP warning activation. After few seconds, LOOP B light also illuminates associated with ENG 1 FIRE warning activation and ENG 1 FIRE handle illumination (and fuel HP valve light illumination if HP valve open). b) LOOP TEST pbRelease c) LOOP A warning and ENG 1 FIRE warning activation disappear. After few seconds LOOP B warning disappears.	

COCKPIT PREPARATION

PF	PM
EVAC SIGNALCAPT	
Usual position is CAPT.	
L/G ANN.....CK	
Normal Indications, Also Cross check with center instrument panel.	
* FUEL PANEL.....CK	
- FUEL X-FEED Check CROSS-LINE - ISOL VALVES/LP VALVES Check in line - Flow bars Check in line and steady NOTE : ISOL VALVES check must be done with “ PED and OVHD PNL” control knob set to BRIGHT.	
* CVR.....TEST	
a) CVR TEST PRESS b) Check correct operation.	
CABIN PRESS.....CK	
- AUTO PRESS RATE LIMIT knob NORM - RATE LT Check extinguished - Check CAB ALT, DIFF PRESS and CABIN V/S for logical indication.	
* CREW OXY.....CK	
- SYSTEM HIGH PRESS check indication between 1400 PSI and 2000 PSI NOTE : If pressure is below 1400 PSI, check oxygen duration chart to determine that quantity is sufficient for the scheduled flight. - SYSTEM LOW PRESS Check in green sector	
ENG 2 FIRE PROTECTION.....CK / TEST	
- ENG 2 FIRE handle.....Check IN and latched - SQUIB TEST pb Press - AGENT SQUIB 1 + 2 lts Check Illuminated - LOOP TEST pb Press and Hold a) LOOP A It comes on associated with LOOP warning activation. After few seconds, LOOP B light also illuminates associated with ENG 2 FIRE warning activation and ENG 2 FIRE handle illumination (and fuel HP valve light illumination if HP valve open). b) LOOP TEST pb release c) LOOP A warning and ENG 2 FIRE warning activation disappear. After few seconds LOOP B warning disappears.	
EMER EXIT LT.....ARM	
Check that DISARM light extinguishes.	

COCKPIT PREPARATION

PF

PM

COMPT TEMP.....11 O'CLOCK

- ECON FLOW ON/AS RQRD
- MAX COOL AS RQRD
- COMPT TEMP Selectors AUTO / 11 o'clock
- Select CRT position.
- **NOTE:** ECON FLOW may be selected ON, if required, For flight with less than 190 pax. (A306), 160 pax.(A310) Approx 68 % flow is provided.

*** FMS INITIALIZATION:**

- **CAUTION:** If the message "PLEASE WAIT" appears, DO NOT PRESS ANY CDU KEY until the message is cleared.

NAVAID DESELECTIONAS RQRD

- If NOTAMS warn of any unreliable DME or VOR/DME, Access NAVAID page and DESELECT the related NAVAID.

F-PLAN INITIALIZATION.....COMPLETE

- INIT Key Press
- INIT A page is displayed
- CO RTE or city pair code Insert
- INIT PAGE  Check / Complete
- Check / modify ALTN / ALTN RTE
- Enter FLT number, For ATC needs, the crew should enter exactly the entire flight number, as shown on the ICAO flight plan, without inserting any space.
- Enter Cost Index as per airline policy:
- Up to 1000 NM trip distance: insert 30.
- Up to 2000 NM trip distance: insert 50.
- Above 2000 NM trip distance: insert 70.
- Enter intended CRZ FL, modify it if necessary taking into account ATC constraints or expected gross weight.
- Enter tropopause altitude and/or modify CRZ FL TEMP according to MET forecast.
- Enter CRZ wind
- The IRS are usually aligned to the departure airport reference point coordinates, however, if the gate coordinates are published, these may be inserted here.

F-PLAN ACOMPLETE

- If CORTE has been inserted, the F-PLN should automatically include the preferential or most probable T/O RWY, approach and landing RWY, associated SIDs, STARs, transition and En Route WPTs. However some data bases will only include departure and arrival airportidents and EN ROUTE WPTs.
- The crew must check, modify or insert (as applicable) the F-PLN according to the data given by ATIS, ATC or MET.
- Lateral revision at Departure Airport, Select RWY then SID then TRANS (if appropriate).
- Lateral revision at WPT for ROUTE modification if needed.
- Vertical revision, check or enter SPD / ALT. CONSTRAINTS according to ATC clearance.
- Check the flight plan versus computerized flight plan and navigation charts.
- Use ND PLAN mode while scrolling the F-PLN on the CDU.

COCKPIT PREPARATION

PF

PM

SEC F-PLAN.....AS RQRD

- This is routinely a copy of the active. However, consideration may be given to the following :
 - Copy the active F-PLN but modify it at a suitable WPT for an immediate return to the departure airfield in the event of, for example engine failure.
 - If weather is below landing minimum at the departure airfield the secondary flight plan should be that required for an immediate diversion after take off.
 - If there is the possibility of a runway and/or SID change during taxi, this can be prepared for by copying the active and making the necessary modifications.

*** FMS DATA INSERTION:**

BLOCK FUEL / ZFW / ZFWCG INSERT
THR RED / ACC ALT SET / CK

AIRCRAFT LIBRARY..... CK

- ZFCG/ZFW Insert
- BLOCK FUEL Insert
- If ZFCG, ZFW or BLOCK FUEL are not yet available, it is acceptable to enter the expected values in order to obtain predictions. (USE OPERATIONAL FLIGHT PLAN).
- If ZFCG and ZFW are inserted, and FUEL PRED prompt is pressed, the FMS will provide the minimum fuel required for the trip according to fuel policy (Route Reserves, ALTN, FINAL).
- If ZFCG, ZFW and BLOCK FUEL are inserted, the FMS will provide all predictions, as well as the EXTRA fuel if any.
- PROG As rqr
- Check VOR tuned by FMS, modify if required.
- Check or modify THR RED/ACC ALT as needed. (Use NADP1 or NADP2 according to regulations).
- TO SHIFT As rqr

• When both pilots seated:

SLIDING WINDOW.....CK

SLIDING WINDOW..... CK

• GLARESHIELD

* PFD / ND BRIGHTNESS AS RQRD
* FPV / FD ON FD
* VOR / NAV / ILS AS RQRD

* PFD / ND BRIGHTNESS..... AS RQRD
* FPV / FD ON FD
* VOR / NAV / ILS AS RQRD

- NOTE :** VOR positions: manual tune. NAV/ILS positions: auto tune.

* EFIS CTL PNL AS RQRD
* ND MODE / RNG AS RQRD

* EFIS CTL PNL..... AS RQRD
* ND MODE / RNG AS RQRD

- NOTE:** It is recommended to use MAP mode for Take-off, unless the SID is not in the data base, then PF should use, ARC or ROSE.
- NOTE:** It is recommended to set the minimum RANGE to display the first waypoint on departure, or as required for weather radar.

* FCU CK / SET

* FCU CK / SET

- Adjust brightness.
- ALT Sel Initial cleared ALT.

COCKPIT PREPARATION

PF

PM

• **LATERAL CONSOLE:**

* RAT..... SAFETY WIRED
OXY MASK100% & TEST

RAT..... SAFETY WIRED
OXY MASK 100% & TEST

- **WARNING:** To prevent ears injuries to ground mechanics connected to the intercom system, inform them that a loud noise may be heard in the headset when performing this test.
- **OXY MASK TEST:**
 - Depress INT on audio panel and adjust volume.
 - Select interphone transmit.
 - Push down and hold RESET/TEST slide control and observe blinker momentarily yellow.
 - Notice oxygen flow sounds through loudspeaker.
 - Release RESET/TEST slide control.
 - Set N/100% manual control to 100%.
 - Check system Low Press indication within green band.

GND SERVICE INTPH.....OFF (CM1)

SNIFFER FAN.....CK (CM2)

* EGPWSNORM

- Check no FAULT light illuminated.
- **NOTE:** EGPWS FAULT light will be illuminated whilst IRS 1 is in align mode if a EGPWS MKII is fitted.

INST SWTG.....NORM

INST SWTG NORM

• **CM1 / CM2 INST PANELS:**

* PFD / ND CK

* PFD / ND CK

- Check ND for correct display.
- Cross check ND and RMI heading with standby compass. If ND/RMI and compass headings disagree :
- MSU mode selector OFF
- After at least 15 seconds :
- MSU mode selector NAV
- ALIGN MODE annunciator Check steady amber
- Upon completion of the 10-minute alignment period:
- Cross-check IRS (ND and RMI) and standby compass magnetic heading indications.
- Check ground speed, initial waypoint, VOR indications.
- Check PFD for correct display.
- Check IAS, FMA, initial target ALT, altimeter readings, attitude.

* STBY ASI..... CK

* STBY ASI CK

- Check speed pointer 0.

* VSI..... CK

* VSI CK

- Check no flag and pointer 0.

* AIRFIELD QNH OBTAIN & SET
* ALTIMETER..... CK

* AIRFIELD QNH OBTAIN & SET
* ALTIMETER..... CK

- Altimeters Check no flag.

COCKPIT PREPARATION

PF	PM
EGPWS TEST	
CLOCKS CK / ADJ	CLOCKS CK / ADJ
- Check time. Adjust if necessary. Set elapsed time and chrono to zero. CTR INST PANEL:	
* STBY ALT CK	
- Check no flag. - QNH Set	
* STBY HORIZON CK	
- Check no flag. - Erect if necessary.	
ALTN BRK CK BRK - A / SKID NORM / ON * OIL QTY CK / BUGS SET * ENG INST CK	
- OIL PRESS Check zero - OIL QTY Check NOTE: MIN OIL quantity before start is 14 QT (A306), 10 QT (A310) + estimated consumption. (estimated consumption is based on each operator's oil consumption monitoring). - OIL QTY bugs Set - FF ind. Check zero - N2 ind. needles and counter Check zero No flags, and red max pointers in front of the red limit marks. - EGT ind. needles and counter Check logical temperature, No flags and red max pointers in front of the red limit marks. - N1 ind. needles and counter Check zero No flags, red max pointers in front of the red limit mark and N1 mode select knobs are pushed and counter for manual setting is masked.	
* REV & REV UNLK LTS EXTINGUISHED * TRP CK	
- TOGA/FLEX Set - TAT Check logical - THR LIMIT bugs Check correspond to THR LIMIT display	
L/G POS DET SYS AS RQRD L/G WARN TEST PERFORM	
- WARN TEST pb Press Observe down arrow illuminates, continuous repetitive chime sounds and ECAM system activates. - POSITION DET SYS1 or 2 Select, Select SYS1 on odd days, SYS2 on even days.	
LDG ELEV SET DEP FIELD ELEV	

COCKPIT PREPARATION

PF

PM

• PEDESTAL:

* ADFsSET
WX RADAR TEST
ECAM RCL
ECAM SYS PAGES REVIEW
* ATC XPNDR / TCASTEST / ABV
* RADIOSSET
* VOR / ILSSET

* ADFsSET
WX RADAR TEST

* RADIOSSET
* VOR / ILSSET

- ADF's Set.
- WEATHER RADAR Test pattern check.
a) Adjust the tilt downward until ground returns appear and then slowly adjust it in 1 to 2 degrees steps, up to 15° UP, for weather returns. Select tilt at 4° UP for takeoff.
- INT knob Check volume Make sure that the knob is turned up to enable ground crew contact.
- VHF COMM Check VHF transmission and reception. Normally VHF 1 is set for ATC, VHF 2 is used for ATIS & Handling agents, VHF 3 is used for 121.50.
- HF Check HF transmission and reception if required for flight.
- Set ATC XPNDR to STBY, MODE "C" ON, TCAS STBY & ABOVE.
- Set ILS frequency for departure runway.

* FMS DATA CONFIRMATION:

INIT PAGE A & B CK
F-PLN A & B COMPLETE / CK
T.O. DATA CALCULATE / INSERT

AIRFIELD DATAOBTAIN / RECORD
INIT PAGE A & BCK
F-PLN A & B COMPLETE / CK
T.O. DATA CALCULATE / INSERT

- PF enters wind/temp for waypoints as provided on the computerized flight plan. Entries should be made at waypoints when there is a difference of either 30 degrees or 30 KTS for the wind data and degrees C° for temperature deviation. It should be remembered that the more points entered, the better the accuracy of the FMS predictions.
- PF reads, distances between waypoints, PM checks the distances as provided on the computerized flight plan.
- Check Distance to DEST along the F-PLN indicated on PROGRESS page. Compare with the computerized flight plan.
- V1, VR Insert.
- SPD/MACH setting knob turn to select V2.
- SPD/MACH setting knob Push to Activate Preset.
- SPD/MACH setting knob Turn to Preselect initial climb speed.
- Airfield data: Obtain necessary data for system initialization and cockpit preparation. This should include RUNWAY IN USE, ALTIMETER SETTING and WEATHER DATA. ATC Tower temperature should be used for all Take-off calculations.
- PM checks FMS data, calculates and prepares take off DATA and bug card.
- **FUEL:**

* FUEL QTYCK

* FUEL QTYCK

- After the refueling is completed :
- FUEL PUMPS NORM
- Check fuel quantity corresponds to FLIGHT PLAN fuel, Crosscheck with sum of individual tank quantities.
- Crosscheck that the sum of the Fuel On Board recorded at the end of the last flight and the fuel uplift (if any) is consistent with the current Fuel On Board. If an abnormal discrepancy is found, maintenance action is due.
- Check fuel load asymmetry within limits.
- **NOTE** : If a dash is displayed in the last digit of any fuel tank indicators, the quantity indication accuracy is fully serviceable.

COCKPIT PREPARATION

PF

PM

TAKEOFF BRIEFING

The purpose of the takeoff briefing is for the **CM1** to inform the **CM2** of the planned course of action for **abnormal situations during takeoff** and for the **PF** to review with the **PM**, the **planned departure procedure**. It is also a recall of **Standard Emergency Procedures** and is the time for the **CM1** to give any specific instructions to the **CM2**.

It should be completed **prior to engine start**.

It consists of **three (3)** subjects, **Emergency Briefing**, **E/O SID OR Routing** and **Departure Briefing**.

It is most important that the takeoff briefing is given at a time when the **cockpit is quiet, workload is low & without the presence of non-essential personnel**, so that both pilots may concentrate on its contents and at a time when the takeoff conditions are likely to be known, and should be completed **not later than 5 minutes** before STD at the end of the Cockpit Preparation.

Takeoff Briefing

Key Points

→→ Nominate The Pilot Flying (**PF**).

Review:

→→ Actions & task-sharing in the event of failures (**1st flight of the day**), for subsequent takeoffs of the day, just mention

► **Standard Emergency Briefing** ◀

→→ Use of anti-ice & bleeds.

→→ Deferred Defect Items (MEL items) & NOTAMS affecting takeoff.

→→ Speeds / Power Setting / Configuration / Trim (use CDU / PFD / Bug-Card).

→→ Taxi-out routing, runway state & surface wind.

→→ Takeoff limitations.

→→ Routing in the event of failures e.g. E/O SID / routing.

→→ Remainder of the briefing regarding SID, departure routing & any special procedure (use CDU / FMA / ND) / radio set-ups / use of WX-radar / EGPWS / ignition etc.

TAKE-OFF EMERGENCY BRIEFING (CM1)

PF

PM

EMERGENCY BRIEFING ► Always by the **CM1**

This will be a **L/H (R/H)** seat take-off.

If any malfunction occurs **before V1**, I will call "**STOP**" or "**GO**".

If the call is "**STOP**", I will apply Maximum Reverser and bring the aircraft to a full stop on the runway. You will cancel the aural warning, monitor deceleration and make **standard call-outs**.

After parking brake is set ON.

I will alert the cabin crew. **You** will inform the ATC.

Then we shall go through the ECAM actions and / or appropriate checklist on my command. We will not vacate the runway until we are certain that evacuation is not required and it is safe to do so. If evacuation is required, it will be done on the runway.

If the call is "**GO**" or malfunction occurs **after V1**, we will continue the take-off. No action will be taken **below 400 ft AAL** except silencing aural warning, raising the landing gear (if applicable) and application of **TOGA** if required. Once safely established in climb and **above 400 ft AAL** you (I) will carry out ECAM actions and/or appropriate checklist on my (your) command up to second agent discharge, if necessary, for an engine fire. **Conditions permitting**, we may take action **bellow 400 ft AAL** in case of emergency situation on my command.

We will assess the situation whether it is possible to continue, return or divert.

In case of immediate return we will be below / above maximum Landing Weight

Limit and if applicable we have to comply with the overweight landing checklist.

E/O SID / Routing will be

Continuation ► **Departure briefing** ► **By PF**

BEFORE PUSH BACK OR START

PF

PM

LOADSHEET.....CK (CM1)

- The Captain should thoroughly check the Load and Trim Sheet (LTS), particularly for gross errors.
- Make sure, that the loadsheet data is correct, e.g. correct flight, correct aircraft, dry operating index, Configuration.
- Check fuel on board and fuel index correction. For aircraft with a trim tank, check the trim tank quantity is as expected on the loadsheet, if not use the actual trim tank quantity to calculate the fuel index correction.
- Check the take off trim setting and that it is optimized for the takeoff weight and flap setting.
- Check the loadsheet CG, against the ECAM CG. In case there is a discrepancy of more than 1.5 %, check that the ZFW and ZFCG have been correctly inserted in the CDU, then rely on the ECAM CG. If there is less than 1.5 %, no further action is required. Rely on the ECAM CG.
- Check that the takeoff CG is within the LTS operational limits.

T.O. DATA CK / REVISE

BUG CARD.....PREPARE

- TAKEOFF DATA Prepare and Check/revise Once the load sheet is checked.
- Check/modify FMS weights/CG
- PM checks or recomputes the takeoff speeds and Flex Temperature using the RTOLW charts.
- PF then INDEPENDENTLY calculates the takeoff speeds and Flex Temperature as a crosscheck.
- Ensure particular care is taken to determine the takeoff configuration.
- Confirm takeoff weight limitation.
- To extend engine life and save maintenance costs, the use of flaps configuration giving the highest flex temperature is recommended. However, when the difference, in term of FLEX temperature, between 2 configurations is low (less than 5 degrees), the highest takeoff configuration is preferable.
- Enter the Flex Temperature and select FLEX on the TRP unless TOGA is required.
- Set V2 and green dot speed on standby ASI (all remaining bugs at 11 o'clock).
- PF enters (or revises) the take off data in the FMS CDU TAKE-OFF and INIT pages.
- Set V2 and Preset initial climb speed on FCU.
- Check CRZ FL is not above MAX FL

SEAT / RUDDER PEDALS ADJ

SEAT / RUDDER PEDALSADJ

- SEAT, SEAT BELTS, HARNESSES, RUDDER PEDALS Adjust
 - a) The seat is correctly adjusted when the pilot's eyes are in line with the red and white balls.

CDU TAKE-OFF

CDUF-PLN

FCU ALT / HDG / SPD.....SET
 PROF..... ARM
 NAV ARM
 APUSTART
 SEATBELT SIGN.....ON
 EXT PWROFF / DISC
 FUEL PUMPS.....ON

ATC CLEARANCE OBTAIN
 ATC CODE SET

- Start APU, If not started before,
- EXT PWR CK OFF, Request external power removal.

BEFORE START C/L DN TO THE LINE

ATC XPNDR..... MODE "S"
 WINDOWSCLOSED
 BEACON ON
 PARK BRK.....CK ON

PUSH BACK / START CLEARANCE OBTAIN
 WINDOWS CLOSED

BEFORE PUSH BACK OR START

PF

PM

- At this stage, the « before start » preparation is stopped to obtain ground crew clearance (For pushback, chocks removed, nose wheel steering selector and by-pass pin in place, tow truck connected and area clear for start/pushback). Obtain ATC pushback and start up clearance.
- Engines may be started during pushback.
- Check cockpit windows closed and locked. Check proper locking of sliding windows by pushing the window handle forward into full closed position.
- Check on ECAM DOOR page that all doors are closed.
- A test of the cockpit door locking system must be done once a day.
- Check that the cockpit door is closed and locked (no cockpit door fault/open indication).
- With the cockpit door spring loaded toggle switch at NORM, the cockpit door is closed and locked.
- For cockpit access request, refer to "procedures and techniques" chapter, "doors" section.

BEFORE START C/L BLW THE LINE

- If no pushback required, check parking brake is on and check brake pressure indication. Chocks may be removed.
- CAUTION:** If, during engine start with parking brake on, the aircraft starts to move due to a parking brake failure, immediately release the parking brake and restore braking using the pedals.
- If pushback required, release the parking brake.
- CAUTION:** Do not use brakes during pushback unless required due to an emergency.
- CAUTION:** Do not use the nose wheel steering during pushback.
- Once pushback is completed apply the parking brake and inform ground crew.

ENGINE START

PF

PM

THROTTLE LEVERS.....CK IDLE
START PRESS.....CK
"START ENG 2/1".....ORDER

ANNOUNCE "N2"

FUEL LEVER.....ON
CLOCKSTART
ANNOUNCE..... "FUEL ON"
ANNOUNCEFUEL FLOW
ANNOUNCE "EGT"
CLOCKRESET
ANNOUNCE "N1"
ANNOUNCE..... "N2 45%"
ANNOUNCE.....(PEAK EGT)

IGN..... START A/B
START 2/1.....PRESS
ANNOUNCE "VALVE OPEN"

ANNOUNCE "OIL PRESSURE"

CLOCK START

CLOCK RESET
ANNOUNCE "VALVE CLOSED"

- Be aware of prevailing surface wind, as tail wind may result in starting difficulties.
- Set IGNITION to START A (odd days) or START B (even days) and check both ARM lights illuminate on START 1 and 2 pushbuttons.
- Note:** PACK VALVE FAULT It illuminates during pack closure.
- ENG 2 is normally started first.
- Announce " START ENGINE 2/1"
- START 2 (1) Press
- Check ARM white light extinguishes, OPEN blue light illuminates.
- Announce "VALVE OPEN"
- BLEED PRESSURE Check increasing.
- When N2 increases:
- Announce " N2"
- OIL PRESSURE Check increasing.
- Announce "OIL Pressure"

ENGINE START

PF

PM

- At 15% N2 Minimum:
 - a) FUEL LEVER ON
 - b) Confirm fuel lever ON and locked/seated in notch.
 - c) Announce "FUEL ON"
- CLOCK Start, 25 sec. max from FUEL ON to light up.
- **NOTE:** As required (quick turn-around, tail wind), set FUEL LEVER to ON at max motoring speed. Max motoring speed is assumed when there is no significant increase in N2.
- FUELFLOW Check in normal range.
 - a) Announce "Fuel Flow"
- EGT INCREASING:
 - a) Announce "EGT"
 - b) CLOCK Stop/reset
- N1 INCREASING:
 - a) Announce "N1 "
- N2 45%
 - a) Announce " N2 45 %"
 - b) When START 2(1) OPEN light extinguished:
 - c) Announce "VALVE CLOSED"
- **NOTE:** Start valve is fully closed when engine reaches ground idle.
- **AT IDLE:** N1, N2, EGT and FF Check.
- At ISA, sea level typical values: N1 about 24%, N2 about 65%, EGT about 400 C and FF about 600 kg/h).

REPEAT THE START SEQ ENG 1/2

BOTH ENGS PRIMARY & SECONDARY PARAMETERS CK
ANNOUNCE..... "ALL PARAMETERS CORRECT"

- Announce "START ENGINE 1/2"
 - a) Repeat the start sequence as stated above.
- **NOTE:** Both ARM lights illuminate when ENG 1 & 2 N2 reach 45%.
- AFTER SECOND ENGINE START:
 - a) ECAM Check Secondary Parameters.
 - b) Announce "ALL PARAMETERS CORRECT"
 - c) OIL QTYCHECK BELOW BUG OR BELOW 20 QUARTS. (At least 30 seconds after the engine reaches stabilized idle and prior to any thrust application for break-away).
- **CAUTION:**
- 1. Monitor N2, EGT and FF indicators closely during the start for any abnormal indications. Sluggish N2 acceleration is an indication of an impending:
 - a) HOT START, if FF is high and EGT rises rapidly.
 - b) HUNG START, if FF is low and EGT is proportionally low.
- 2. The start attempt should be discontinued if:
 - a) EGT does not rise within 25 seconds after FUEL ON.
 - b) Sluggish N2 acceleration accompanied by rapidly increasing EGT or/and tail pipe torching reported by ground crew.
 - c) N2 stabilizes below ground idle.
 - d) If fuel or ignition is inadvertently interrupted. After placing the FUEL lever to OFF, maintain starter engagement and continue motoring the engine for 30 seconds to ventilate the combustion chambers.
- 3. The start attempt should be discontinued if an indication of N1 is not obtained within 30 seconds of N2 idle operation.
- 4. If starter engagement is interrupted, the starter may be reengaged below 20 % N2. (30 % maximum, 0 % recommended).
- 5. Operational conditions may necessitate the use of starter inlet air pressure below 25PSI. Should these conditions arise, extended time to idle and higher EGT peaks may result. Under these conditions, the start cycle must be closely monitored and appropriate action taken to prevent a HUNG or HOT START.

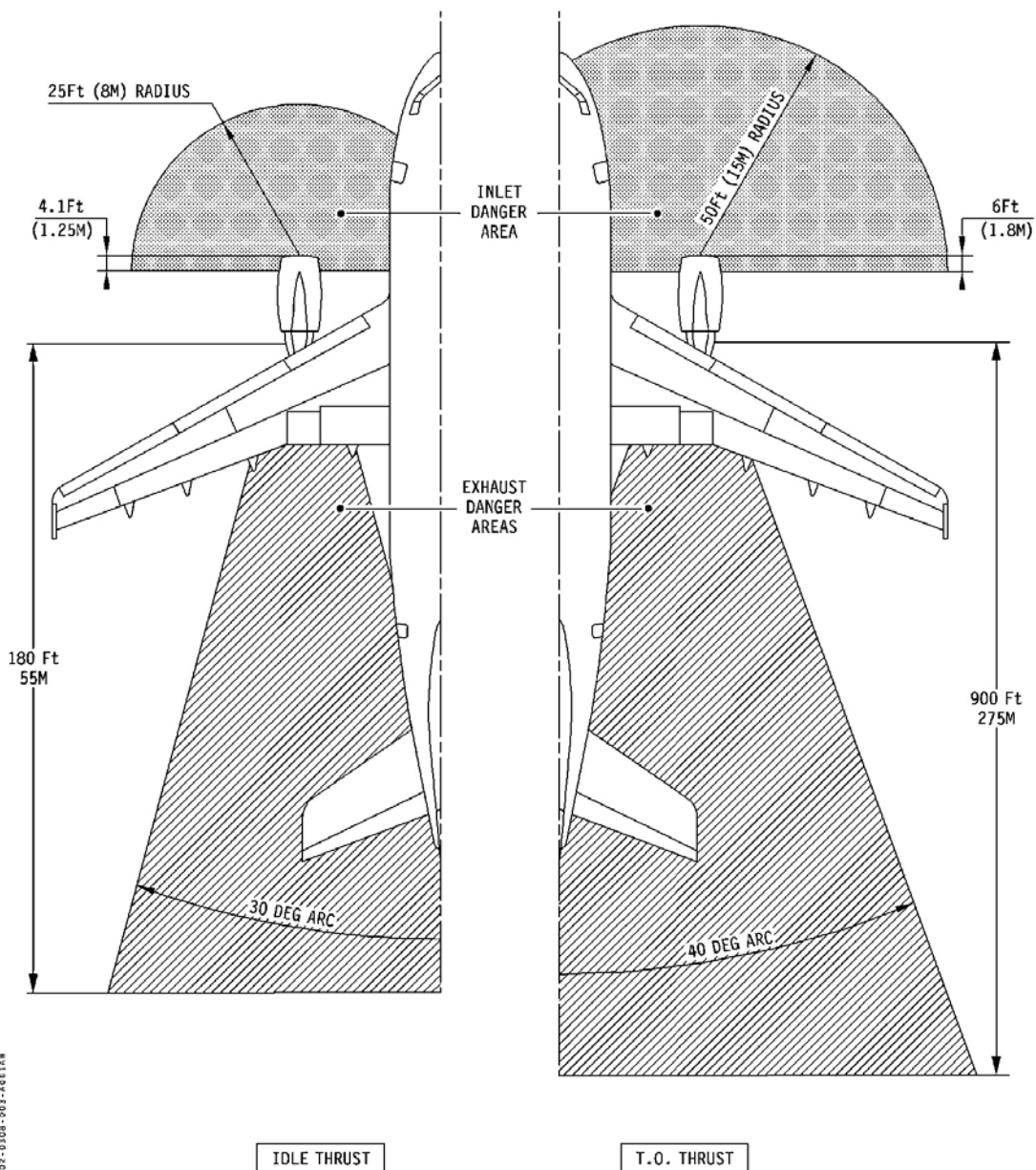
ENGINE START

PF

PM

GROUND RUN UP - DANGER AREAS

GROUND RUN UP - DANGER AREAS



60FC-02-0308-003-406148

AFTER START	
PF	PM
APU BLEED.....OFF APU OFF IGN OFF	
• APU BLEED is selected OFF just after engine start to avoid engine exhaust gas ingestion. • APU MASTER SW OFF • NOTE : APU shut down may be delayed if APU BLEED supply is required. • IGNITION..... OFF a) ARM lights extinguish. b) The PF turning the IGNITION rotary selector to OFF is the signal for the PM to commence the AFTER START actions. • CAUTION: After start to avoid thermal shock, the engine should be operated at idle or near idle for at least 3 minutes prior to advancing the thrust lever to high power. Taxi time at idle may be included in the warm-up period.	
WING ANTI ICEAS RQRD ENG ANTI ICEAS RQRD	
• WING ANTI ICE AS RQRD • NOTE : WING ANT/ICE valves stay closed as long as the aircraft is on the ground. • ENGINE ANTI ICE.....AS RQRD • NOTE: Icing conditions may be expected when OAT or TAT is below +8C, with visible moisture. • NOTE: During ground operation in icing conditions, periodic engine run-ups must be performed to shed any possible ice build-up from the spinner, the fan blades or low pressure compressor stators. Periodic engine run-ups must be performed, every 30 minutes, by accelerating the engine to 60 % N1 minimum for 30 seconds (or as practical depending on traffic and airport conditions). Before takeoff a last engine run-up must be performed with observation of all engine parameters to ensure normal engine operation.	
LDG LTS.....EXTEND GND SPLRS.....ARM AIL / RUD TRIMZERO ECAM.....CK STS ECAM DOOR PAGE.....CK CLEAR TO DISCONNECTORDER GND CREW CLEARANCE.....OBTAIN	BRK FAN.....ON SLATS / FLAPSSET PITCH TRIMSET GND CREW CLEARANCE.....OBTAIN
AFTER START C/L	
• SLATS and FLAPS..... SET for T/O a) Set SLATS/FLAPS and KRUGER for T/O, and check position on SFPI. b) If the OAT is below - 40° C extend the SLATS/FLAPS step by step. c) If there is snow or slush on the taxiways, maintain the slats/flaps retracted until reaching the holding point before takeoff. • BRAKE FAN is selected ON, regardless of OAT. • NOTE: If the OAT is below - 40°C, Check correct operation of the electrical pitch trim prior to setting the PITCH TRIM for T/O. • ECAM Check STATUS • ECAM DOOR Page Select a) Check all slides armed. b) Deselect DOOR page after slide verification. • Clear to disconnect Announce a) Ground crew must ensure chocks removed, nose wheel steering by-pass pin removed, area clear for taxi and interphone is disconnected. b) Check for hand signal display on the left/right side to show all clear for taxi. • AFTER START CHECK-LIST Completed	

AFTER START

PF	PM
FLT CTL CK ANNOUNCE "NORM FORT.O."	FLT CTL CK T.O. CONFIG TEST AND ANNOUNCE
- ECAM Select F/CTL page - FLIGHT CONTROLS Check a) The PM checks full travel and feel of the elevators and ailerons and spoilers, whilst monitoring on the ECAM F/CTL page. b) The PF holds the nose wheel steering handle to maintain the aircraft direction and checks full travel and feel of the rudder, whilst the PM monitors the ECAM F/CTL page. Check all indications return to zero position with respective controls at neutral, except the ailerons where droop position is indicated. c) If the OAT is below- 40° C, the PM should check full extension of the speed brakes by setting the speed brake lever notch by notch up to the FULL position whilst monitoring ECAM F/CTL page. Note: HYD SYS LO PR warning may occur if test is performed on more than one axis at a time. Note: FLIGHT CONTROL check should be done with CWS OFF. If CWS is selected full travel will not be available. Note: During aileron check, rudder deflects left or right coordinating with aileron movement (YAW DAMPER input).	

TAXI

PF	PM
ELAPSED TIME START EXT LTS SET PARK BRK RELEASE BRAKES CK	TAXI CLEARANCE OBTAIN ELAPSED TIME START ANNOUNCE "PRESSURE ZERO"
- TAXI CLEARANCE Obtained - ELAPSED TIME Start a) To record block time. - EXTERIOR LIGHTS SET - Select NOSE Light to TAXI, WING and RUNWAY TURNOFF lights, ON, Day and Night. - PARKING BRAKE OFF a) When cleared to taxi, the PF should set the PARK BRK handle to OFF, while holding brake pedals pressed. When the aircraft starts to move, the PF should check the efficiency of the normal braking system by smoothly pressing the brake pedals, to ensure that the aircraft slows down and call "BRAKE CHECK". The PM should also check the brake pressure triple indicator to ensure that brake pressure remains at zero and responses "pressure zero". This indicates a successful changeover to the normal braking system. This indicates that there is no residual yellow pressure. b) Any indicated pressure below approximately one needle width on the brake pressure triple indicator can be considered as zero. WARNING: If the breaks fail during ground operations, immediately select the BRK/A/SKID sw to ALTN-OFF and modulate the breaks with pedals. break pedals should be released when the A/SKID is switched OFF. otherwise the pedal breaking orders will be taken into account and the aircraft will react strongly. in an extreme emergency and only if pedals are ineffective with the antiskid OFF the aircraft may be stopped with the parking break (full pressure application will occur) - THRUST LEVERS As rqrd a) In order to get the aircraft moving, little if any power above idle thrust will be required (max 40% N1). Thrust should normally be used symmetrically. Once aircraft is moving little thrust is required. b) The wing mounted engines are close to the ground. Avoid placing engines over unconsolidated or unprepared ground e.g over the edge of taxiways. c) Avoid high thrust settings at low ground speeds due to the risk of ingestion (FOD).	

TAXI	
PF	PM
- The normal maximum taxi speed should be 30 kt in a straight line, 15 kt for a sharp turn. The ground speed is difficult to assess so monitor ground speed on ND. Do not "ride" the brakes, as 30 kt is exceeded, apply brakes smoothly and decelerate to 10 kt, release the brakes and allow the aircraft to accelerate again. - Below- 40° C, small braking inputs are required during Taxi. XX will be indicated in place of the brake temperatures while the temperatures are below 0°. CAUTION: If the brakes fail during ground operations, immediately select the BRKJA/SKID sw to ALTN-OFF and modulate the brakes with pedals. Brake pedals should be released when the A/SKID is switched OFF. Otherwise the pedal braking orders will be taken into account and the aircraft will react strongly. In an extreme emergency and only if pedals are ineffective with the antiskid OFF the aircraft may be stopped with the parking brake (full pressure application will occur). CAUTION: If aircraft has been parked in wet conditions for a long period, efficiency of first brakes application at low speed will be reduced.	
T.O. DATA / CONDITIONS CK / REVISE FMA CK FLT INST CK TERR ON ND AS RQRD ECAM MEMO PAGE REVIEW TAKE OFF BRIEFING REVIEW	T.O. DATA / CONDITIONS CK / REVISE FMA CK FLT INST CK TERR ON ND AS RQRD
- ATC clearance Obtain/Confirm - TAKEOFF DATA/CONDITIONS Check/Revise - If takeoff data have become more limiting such as wind change or Tower temperature increase, or in case of runway change, prepare updated takeoff data and as appropriate: a) SLAT/FLAP LEVER Select New Takeoff position b) V2 Reset on FCU c) Initial climb speed Preset on FCU d) FLX TO temperature Reinsert in e) TRP PITCH TRIM Check/Adjust If flap setting is changed, pitch trim setting should be adjusted. f) V1, VR Reinsert g) F-PLN (Runway) Revise h) F-PLN (SID, TRANS) Revise or check, Check with ND in MAP mode, with a 15 NM range. Particular care should be taken to confirm the ATC clearance agrees with the FMS if NAV is to be used. i) FCU As rqr, AP Should not be selected in CMD for Takeoff j) CLEARED ALT Set on FCU k) IF PROFILE NAV mode is used: (PROFILE Arm, NAV Arm) l) INITIAL AFTER T.O. HOG Set on FCU/As rqr, If an ATC HDG is required after takeoff set the heading on the FCU, NAV will be disarmed. m) FD Check selected ON n) FMA Check - FLIGHT INSTRUMENTS Check a) Scan instruments panels observe no abnormal flag on instruments. - ATC code Confirm/Set - TERR ON ND As rqr, In mountainous areas, consider displaying terrain on ND. If use of radar is required, consider selecting the radar display on the PF side, and TERR ON ND on the PM side only. - ECAM NORM FOR TO.	

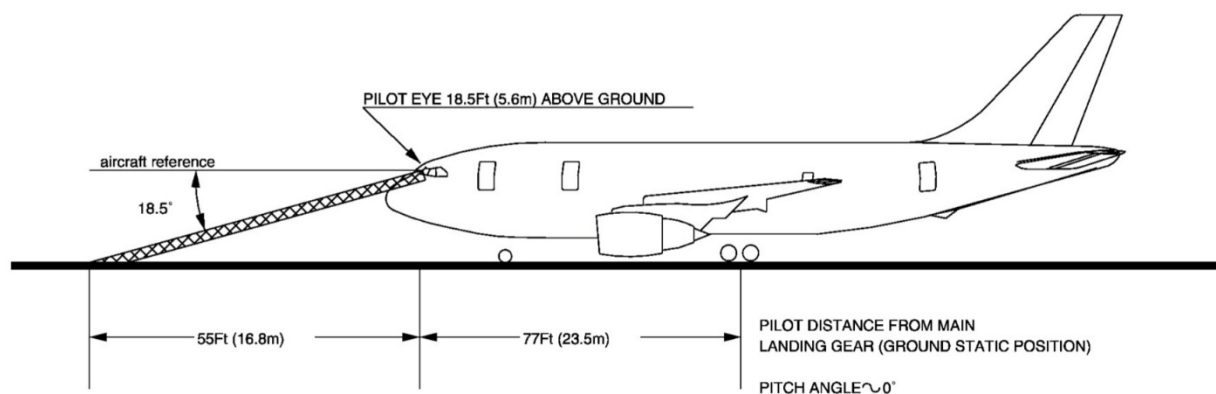
TAXI	
PF	PM
- TAKE OFF BRIEFING Confirm a) This briefing should normally be only a brief confirmation of the thorough takeoff briefing made at the gate. Any changes in the clearance should be addressed at this time. As extensive use as possible of the displays should be made. eg. "Takeoff on RWY 07 (TAKE OFF page), weight 150 t (left ECAM), Slat/Flap 15/15 (SFPI +handle), 50T fuel (L-ECAM), FLEX 50°, 93% N1 (TRP), LMG2D departure (F-PLN page) V1 151, VR 161, V2 164 (TAKE OFF page +PFD), initial cleared ALT (PFD+ FCU)."	
CABIN REPORT.....OBTAIN (CM1)	
Obtain cabin report from purser, as a minimum : « CABIN SECURED for T.O. »	
BEFORE T.O. C/L DOWN TO THE LINE	

TAXI

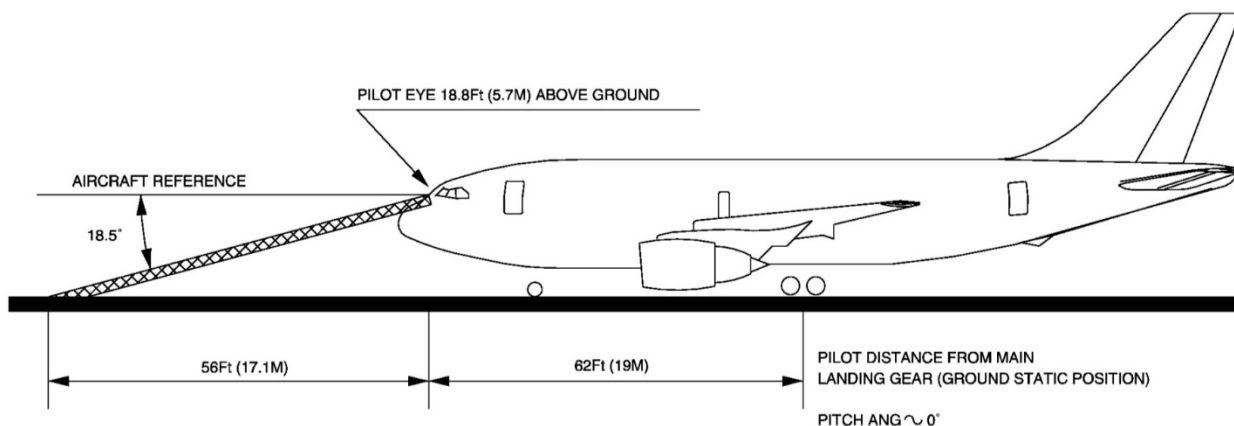
PF

PM

• VISUAL GROUND GEOMETRY A300-600R



• VISUAL GROUND GEOMETRY A310



BEFORE TAKE OFF

PF	PM
BEFORE T.O. PROC.....ORDER STROBE LTS.....ON APP PATH.....CLEAR OF TRAFFIC	T.O. / LINE UP CLEARANCE OBTAIN APP PATH.....CLEAR OF TRAFFIC CABIN CREW.....ADVISE
- TAKEOFF or LINE UP CLEARANCE Obtain - APPROACH PATH CLEAR OF TRAFFIC Check - CABIN CREW Advise; PM announces: "Cabin crew take off position"	
	IGN.....CONT RELIGHT / AS RQRD PACKS OFF
- CONT RELIGHT is advisable if: a) Runway with standing water. b) Heavy rain. c) Heavy rain or severe turbulence expected after takeoff. - If required to increase takeoff performance, select pack valves OFF, or use APU bleed to supply the packs. a) If APU bleed supply is used and wing anti-ice is required one pack must be selected off.	
QFU & THRESHOLD.....CONFIRM	BRK TEMP.....CK BRK FAN.....OFF AUTO BRK.....MAX ATC XPNDR.....MODE "C" TCAS.....TA/RA RADAR.....AS RQRD QFU & THRESHOLD CONFIRM
- BRAKE YELLOW HYDRAULIC PRESSURE Check a) If residual pressure is indicated: Depress several times the brake pedals until release of residual pressure. b) If residual pressure remains : Return to gate. Maintenance action is due. - BRAKE TEMPERATURE Check a) Check brake temperature on ECAM. If brake fans are ON and indicated brake temperatures are above 150°C the takeoff must be delayed. Continue brake fan operation until the indicated temperature drops below 150°C. NOTE: With brake fans ON, a temperature indication of 150°C is equivalent to an actual brake temperature of 300°C. Takeoff is only permitted with actual brake temperature below 300°C. If brake fans are OFF, takeoff may be continued unless the BRAKE TEMP HI caution appears, which is activated at 300°C. - AUTO BRK MAX a) ON It illuminates. b) AUTO BRK may be armed with parking brake on. c) Selection of MAX mode before takeoff will improve safety in the event of an aborted takeoff. If takeoff must be aborted, the autobrake system will apply the maximum braking (if the ground speed is above 85 kt) as soon as the thrust levers are set to idle which represents a single action done without delay. - ATC XPNDR MODE "C" - TCAS Set to TA/RA a) TA ONLY mode should be selected at particular airports and during particular procedures identified by an operator as having a significant potential for unwanted or inappropriate RAs (closely spaced parallel or converging runways...). - QFU and THRESHOLD CONFIRM a) Check and confirm QFU and THRESHOLD versus the airport chart.	
BEFORE T.O. C/L BELOW THE LINE	

TAKE OFF

PF

PM

ALL EXT LTS.....ON
ANNOUNCE....."TAKE OFF"
CHRONO.....START
THRUST40% N1
BRKRELEASE
GO LEVERS.....TRIGGER

CHRONO START

- ALL EXT LTS ON
 - a) Use NOSE TO, RUNWAY TURN OFF, WING and LANDING lights to minimize bird strike hazard during T.O and set NAV and LOGO as required.
 - b) Set STROBE lights to ON before entering the runway.
- PF progressively adjusts and stabilizes engine thrust from idle to about 40 % N1.
- **NOTE:** Intermediate thrust setting will ensure that both engines will further accelerate similarly and will minimize additional rudder pedal corrections for directional control.
- **NOTE:** CM1 keeps his hand on the throttles until V1 is reached without interfering with the throttle movement.
- **NOTE:** In case of asymmetric throttle movement, CM1 is ready to override or to disconnect the A/THR.
- PF applies sufficient forward control column until 80 Kt, then releases progressively to achieve the neutral position by 100 Kt.
- **NOTE:** CG position ,runway and crosswind conditions dictate the necessary forward input. Objective is to counteract the pitch up moment during thrust application and to increase nose wheel adherence.
- **NOTE:** In case of vibrations or shimmy of the nose wheels, release the forward pressure progressively and apply slight back pressure on the control column as necessary. Vibrations will probably disappear as the weight on the nose wheel is reduced.
- **CAUTION:** PF should use rudder pedal steering for directional control. The use of the nose wheel steering would cause over correction and should be avoided.

CM1 Keeps his hand on throttles after GO LEVERS trigger until V1 is reached

PFD/ND.....SCAN
FMA INDICATION.....ANNOUNCE
BELOW 80 Kts :

AT 100 Kts :
ANNOUNCE CK

PFD / ND.....SCAN
N1 (EPR).....CK
ANNOUNCE.....THRUST SET
AIRSPEED/ENG INST.....SCAN
ANNOUNCE....."100 Kts"

- PFD / ND SCAN
 - a) Check FMA on PFD: THR, SRS, RWY (or HDG or HDG/S), FD1 (2)
 - b) Announce FMA indications
 - c) Check FMS position update (AIRCRAFT symbol on runway threshold on ND).
- TAKEOFF N1 CHECK
 - a) Check takeoff N1 is set prior to reaching 80 kt.
 - b) Announce "THRUST SET"
- AIRSPEED and ENGINE INSTRUMENTS SCAN
 - a) Scan the airspeed and engines instruments throughout the takeoff roll.
- When speed is 100 Kt on PFD :
 - a) **PM** Announce "ONE HUNDRED KNOTS"
 - b) **PF** crosscheck speed reading on own PFD.
 - c) **PF** Announce "CHECK"
- **NOTE:** The FMA mode THR changes to blue indication (ATS declutch by 100 Kt.)
- **NOTE:** Below 100 Kt the decision to abort the T.O., may be taken at the discretion of the captain. Above 100 Kt, rejecting the T.O. is a more serious matter and becomes more critical as V1 approaches.

TAKE OFF	
PF	PM
<u>AT V1 :</u> <u>AT VR :</u> <u>AT VSI POSITIVE & RAD ALT INCREASE :</u> ORDER....."GEAR UP" AP.....ON/AS RQRD	ANNOUNCE "V1" ANNOUNCE "ROTATE" ANNOUNCE "POSITIVE CLIMB" L/G.....SELECT UP GND SPLRS.....DISARM NOSE TAXI LTOFF L/G LVR.....NEUTRAL ANNOUNCE "GEAR UP / NEUTRAL"
• ROTATION PERFORM a) At VR, initiate the rotation with a positive control column input to achieve a continuous rotation rate of about 3°/sec, towards a pitch attitude of 15 degrees, (12.5 degrees if one engine is failed). b) Avoid spoiler extension on the ground and during rotation, by minimizing lateral control to that which is required to maintain approximately wings level. c) After lift off, follow the SRS pitch command bar. • CAUTION: High rotation rates may lead to tail strike. • Announce "POSITIVE CLIMB" a) Announce positive climb when the vertical speed indication is positive and the radio altitude has increased. • WING LTS, RWY TURN OFF LTS and LDG LTS will be left ON, until passing F.L. 150. • AP AS rqrD a) AP 1 or AP 2 may be engaged. If AP is already engaged in CWS; press CWS/CMD pushbutton to engage CMD. • L/G lever Neutral a) When lights are extinguished, set the L/G lever to neutral. b) Announce "GEAR UP / NEUTRAL"	
<u>AT THRUST RED ALT :</u> FMA / CL THRUST CK / ORDER <u>AT ACCEL ALT :</u> <u>AT "F" SPEED POSITIVE :</u> ORDER....."FLAPS ZERO" <u>WHEN SFPI SHOWS FLAPS ZERO :</u> <u>AT "S" SPEED POSITIVE :</u> ORDER....."SLATS RETRACT" <u>WHEN SFPI SHOWS SLATS ZERO :</u> AFTER T.O./CLB C/L DN TO THE LINE	ONE PACKON ANNOUNCE "SPD CKD SELECTING FLAPS ZERO" ANNOUNCE "FLAPS ZERO" ANNOUNCE "SPD CKD SELECTING SLATS ZERO" ANNOUNCE "SLATS RETRACTED" 2ND PACK.....ON
• AT THRUST REDUCTION ALTITUDE: a) THROTTLES Check symmetrical retard movement b) ENGINES THRUST Check CL setting c) TRP Check LIM MODE Indication "CL" and "AUTO" • NOTE: In case of either asymmetrical thrust reduction or thrust reduction below CL, A/THR must be disconnected and thrust manually set. • NOTE: With a throttle lever stuck, whenever the ATS commands a thrust change, the free throttle lever can move to an unexpected position leading to a thrust asymmetry and a possible flight path deviation. • Announce FMA indication a) Check P.THR and P.CLB engagement. • PACK VALVE Select one pack a) Check flow bar in-line. • NOTE: Selecting both packs simultaneously may affect passenger comfort. • NOTE: Selecting a pack before reducing T/O thrust may lead to an engine over boost. • Retract FLAPS once above acceleration altitude (or once in CLB phase). This ensures that the aircraft is effectively accelerating towards CLB speed. a) At "F" speed minimum (F on PFD), order "FLAPS ZERO" b) At "S" speed minimum (S on PFD), order "SLATS ZERO"	

TAKE OFF

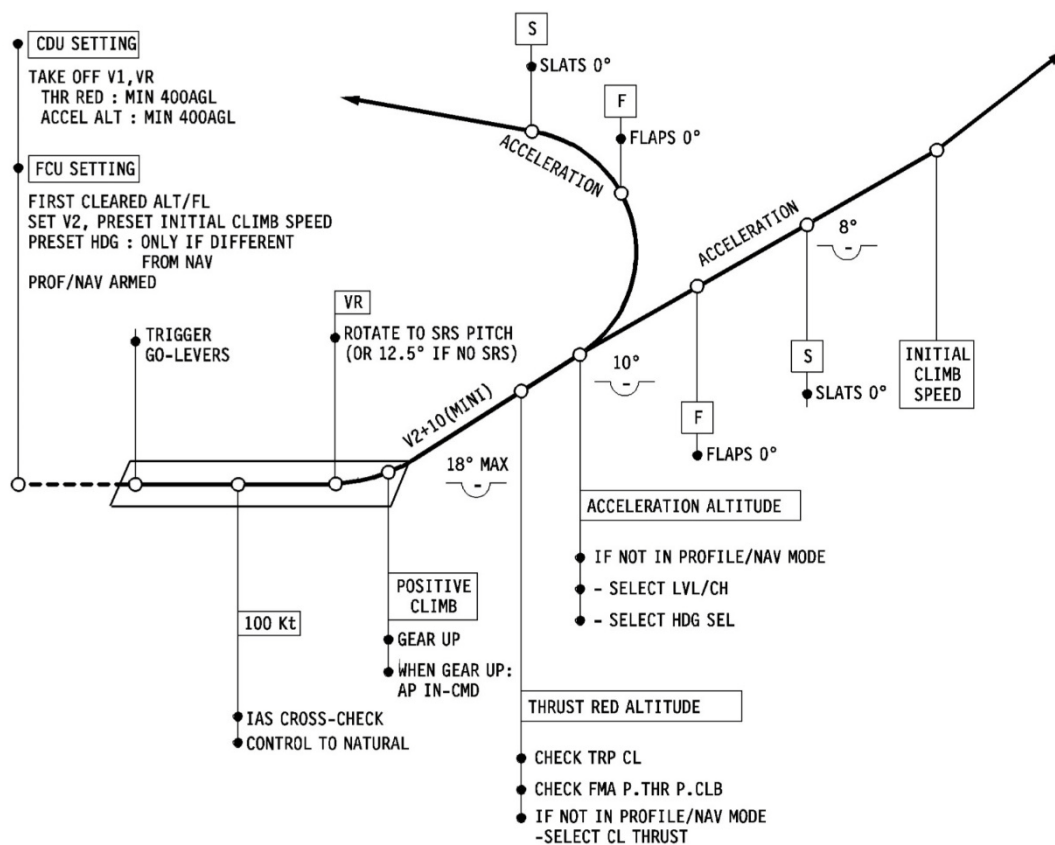
PF

PM

- PACK VALVE Set second to ON
 - a) Check flow bar in-line.

• NORMAL TAKE OFF PATTERN

NORMAL TAKE OFF PATTERN



AFTER TAKE OFF / CLIMB

PF	PM
FMA..... MONITOR / ANNOUNCE ANY CHANGES PF CDU..... PROG FCU / FMS..... SET (IF AP ON) <u>AT TRANSITION ALTITUDE:</u> ALTIMETER STD CRZ F.L..... CK / MODIFY AFTER T.O./CLB C/L BLW THE LINE	FMA..... MONITOR ANY CHANGES PM CDU..... F-PLN FCU / FMS..... SET (IF AP OFF) ALTIMETER STD CABIN TEMP..... MONITOR

- b) **NOTE:** If the APU has been used to supply air conditioning during take off, select APU BLEED to OFF. Use of APU BLEED is limited to 20000 ft.
- c) FMA: The normal climb mode is P.THR/P.CLB in profile mode.
- d) When selecting a new cleared FL:
 - a) FMA indication Check/Announce
 - b) Cleared ALT on PFD Check
- Monitor the progress in climb, particularly when approaching the cleared altitude.
- When 1000 ft below the cleared altitude:
 - a) Announce "One thousand to go"
 - b) Monitor reduction in V/S as the cleared altitude is reached.
 - c) In V/S or LVL!CH mode, monitor ALT* engagement, then ALT at cleared altitude.
- With the AP engaged the PF should make any required F-PLN revisions, But not below FL 100.
- If a speed change is required by ATC, for turbulence or operational reasons (e.g. increase climb rate), select new speed using FMS TACT MODE.
- At transition altitude set standard on all altimeters. Cross check baro settings and altitude readings.
- If ATC clears the aircraft to intended CRZ FL or above, there is no need to modify the CRZ FL inserted in the INIT A page during cockpit preparation. Higher CRZ FL will be taken automatically into account by FCU ALT knob selection.
- **If ATC limits CRZ FL to a lower level than the one inserted in the INIT A page (or present on PROG page) it is necessary to insert this lower CRZ FL in the PROG page. Otherwise there is no transition into CRZ phase : Consequently the speed targets and Mach are not modified.**
- **NOTE:** Pilots should limit vertical speeds to 1500 ft/min during the last 2000 ft of a climb or descent.

F.L. 150 EXT LTS AS RQRD F.L. 200 TERR OFF F.L. 200 MIN..... SEAT BELT SIGN OFF	F.L. 200..... TERR OFF
---	------------------------

CRUISE

PF	PF
• The MAX FL displayed on the PROG page gives at least 0.2g buffet margin. It is not recommended to fly above this level. If a level higher than MAX FL is entered as the CRZ FL, then the message "CRZ FL above MAX FL" will appear on the CDU. • The OPT FL displayed is a function of the Cost Index and aircraft weight. • RADAR: Adjust tilt angle depending on aircraft attitude and the selected range of the ND. A slightly negative tilt is required to avoid over-scanning and to provide some ground returns at the top edge of the ND. • When time permits : a) Recopy the active flight plan in the secondary if an immediate return flight plan has been constructed. b) Check optimum and maximum altitude capability.	

• TOC: TRP (CK) CL OIL QTY BUGS..... RESET TCAS ALL mrofreq..... gnifeirb esiurc	TOC FUEL..... RECORD
--	----------------------

Both pilot should consider DRIFT DOWN PROCEDURES IN CASE OF DECOMPRESSION and DRIFT DOWN PROCEDURE IN CASE OF ENGINE FAILURE during cruise or any time required enroute:

- DRIFT DOWN AND LRC SPD
- ONE ENG GROSS CEILING AND ONE ENG GROSS FLIGHT PATH DESCENT AT DRIFT DOWN SPEED
- TURBULENCE TARGET SPEED AND THRUST SETTING

CRUISE	
PF	PM
Check LIM MODE indication CL (Set CR if not in PROFILE).	
TOC & EVERY 30 MINUTES MIN ECAM MEMO/STS.....REVIEW ECAM SYS PAGES.....REVIEW FLT PROG MONITOR FUEL QTY & DISTRIBUTIONCK ACFT TRIMMINGCK NAV ACCURACYCK	FLT PROG MONITOR / RECORD
FLIGHT PROGRESS: - Periodically review system pages and in particular : - ENG: Oil Press and temperature. - HYD: Fluid quantities. Green sys is lower than on the ground following L/G retraction. - AC DC: GEN parameters, IDG OIL temp, Battery charging status. - BLEED: Bleed parameters. - COND: Check duct temperature compared with zone temperature. Avoid large difference to improve pax comfort. - PRESS: Check Cabin ALT and V/S. - FUEL: Check distribution. - F/CTL: Note any unusual surface position. - OIL OTY MONITOR - Flight progress should be monitored in the conventional way. - When overflying a waypoint; Check the track and distance to the next waypoint. - When overflying a waypoint, and every 30 minutes: - Check Fuel: check FOB, Fuel PRED (FMS) and compare with computed Flight Plan or in cruise quick check table. - Check correct fuel feeding. - Check that the sum of the Fuel On Board and the Fuel Used is consistent with the Fuel On Board at departure. If the sum is significantly greater than the Fuel On Board at engine start, suspect Fuel Quantity Indication over-read. If the sum is significantly less than the Fuel On Board at engine start or decreases, suspect a fuel leak. CAUTION: This check must also be performed each time a fuel imbalance procedure is necessary. Perform the check before applying the fuel imbalance procedure. If fuel leak is confirmed, apply the fuel leak procedure. - Periodically check the OPT and MAX FL to ensure that the aircraft flies as close to the OPT FL as practical. Allowing the CRZ FL to become significantly below the OPT FL may result in a large increase in fuel burn. Check the CRZ FL against that expected in the flight plan, on which the fuel requirements were based. AIRCRAFT TRIMMING: The minimum drag for cruise flight is obtained, when the control wheel is neutral. This condition is obtained by the following procedure: - Ensure symmetrical fuel loading. - Ensure accurate symmetrical thrust, autothrottle disengaged. - Engage the autopilot, if not already engaged, in HDG SEL mode and in ALT mode in CMD. - Adjust the rudder trim in order to get a zero control wheel position (aileron deflection scale on the wheel), between 1.5° L and 1.5° R in a typical normal range. - Verify the bank angle is not too large for passenger comfort (1.5° appears to be a reasonable value). - Check again the lateral trim conditions and retrim if necessary when there is a noticeable change in flight conditions. - Revert to PROF/NAV mode, as required once trimming is completed. NOTE: the same procedure applies to the low speed range of the aircraft including single engine climb out.	

CRUISE

PF

PM

- **Navigation accuracy** must be monitored, particularly when "I" (IRS only) or "LOW" is displayed on the PROG page.
 - a) Insert a VOR/DME ident in the FMS BRG/DIST TO function, check the bearing and distance compared to the same VOR/DME remotely tuned on the PROG page and displayed on the DDRMI.
 - b) The check is satisfactory if the difference in the distance is:
 - 3 NM en route
 - 2 NM in TMA
 - 1 NM on approach
- If the check is satisfactory FMS position is reliable:
 - a) ND ARC or MAP and FMS NAV MODE may be used.
- If the check is unsatisfactory FMS position is not reliable:
 - a) Refer to raw data for navigation and monitor.
- If a gross mismatch between display and real position is detected : disengage NAV mode and use raw data navigation (possibly switch to ROSE MODE so as not to be misled by FMS data).
- Switch off the enhanced functions of the EGPWS via the TERR MODE pushbutton switch located on the Captain's switching panel.
- **RADAR TILT** Adjust
 - a) A near zero degree tilt setting should be used up to around 20, 000 ft. If two different ranges are selected on the two NDs, it is recommended to set a downward tilt with the shorter ND range in order to monitor and detect nearby weather activity. A near zero tilt is recommended for the longer range ND in order to identify weather requiring a possible change of course.
 - b) At higher altitudes above 20, 000 ft, a slightly downward tilt is recommended for both NDs.
- **CABIN TEMP**, Regular attention should be paid to the ECAM CRUISE page so that passenger cabin temperatures may be monitored and adjusted as necessary.
- **NOTE:** It will take up to 20 minutes for the cabin temperature to stabilize after the adjustment of the COMPT TEMP selectors, so do not chase the ECAM cabin temperature indications. Make one change to the COMPT TEMP selectors then wait for the temperature to stabilize prior to any further adjustments.
- **NOTE:** If the aft galley area temperature becomes too low, the most efficient way to increase the temperature, without making it too hot in the passenger cabin, is to increase the bulk cargo heating (if installed) as long as this is compatible with the load carried.

DESCENT PREPARATION

PF

ECAM MEMO / STSCK
LDG ELEVSET
FUEL CK
FMSPREPARE
NAV AIDS SET
EGPWS LDG SLATS / FLAPS SW AS RQRD
TCASBELOW
APP BRIEFINGPERFORM
CABIN CREWADVISE

PM

WX / LDG INFOOBTAIN
BUG CARDPREPARE

- **Descent Preparation** and approach briefing take approx. 10 minutes. So they should be initiated approx. 80-100 miles before top of descent.
- Check STATUS on MEMO page. Review if required. Take particular note of any landing capability downgrade or any other aspect affecting the approach and landing.
- Check weather at alternate and destination, including runway in use.
- **LANDING PERFORMANCE CONFIRM:** Perform an in-flight landing performance assessment if the landing conditions changed compared with the landing distance computation at dispatch, or with a previous computation (e.g. runway, weather conditions, in-flight failure affecting performance, diversion).
- **Note: 1.** If the weather conditions are expected to change at the landing airport, or in the event of significant precipitation, the flight crew should consider a second calculation of the in-flight landing distance with the worst possible runway condition.
- **Note: 2.** The selection of REV MAX is the standard practice for landing. However, on DRY runways the flight crew may select REV IDLE.
- On WET runways (runway surface condition GOOD), the flight crew may select REV IDLE, if all of the following conditions are satisfied:
 - A- A landing distance assessment has been made with the following parameters:
 - MEDIUM TO POOR landing performance level for the in-flight landing distance computation
 - No reverser credit
 - B- The result of this in-flight landing distance assessment is within the LDA.
- **Note:** In case the destination field elevation is higher than the actual cruise CAB ALT, combined with low aircraft cruise FL, set the LANDING ELEVATION counter to landing field elevation before initiating the descent, in order to permit CAB ALT to reach the landing field elevation before landing.
- If flight has been performed below FL 200, check on ECAM FUEL page that there is less than 2000 kg (4400 lbs) of fuel in trim tank. If there is more than 2000 kg (4400 lbs) of fuel in trim tank, check TRIM TK PUMPS are selected ON and select TRIM TK MODE push-button to FWD position.
- Set speed bugs on STBY ASI, VAPP and Green Dot.
- On FMS APPR page:
 - a) LANDING CONFIG Check
 - b) If landing in 15/20 config, select 15/20 on FMS APPR page.
 - c) MDA Insert
- **NOTE:** Some authorities may require operators to add a certain number of feet to the MDA. (In this case add 50 feet).
- DESCENT WIND PROFILE INSERT
 - a) This insertion, on the DES FORECAST page should be made early to ensure optimum Top of Descent point can be re-computed and ensure that it is ahead of the present position.
- STAR/APPROACH Insert
 - a) The FMS computes VAPP based on the predicted landing weight, it is equal to VREF + WIND CORR (entered by crew) + 5 kt (This assumes A/THR is engaged for landing).
- The DECEL point, where the aircraft should start to decelerate is computed based on the inserted flight plan route and wind profile.
- NAVAIDS Check
 - a) Set NAVAIDS as required and check idents. If a VOR/DME exists close to the airfield, it should be selected systematically and its ident should be set on PROG page BRG/DIST TO field for navigation accuracy monitoring during descent.
- GO AROUND page Check/Modify
 - a) Check/modify the THR RED ALT and ACC ALT, Based on NADP1 or NADP2.

DESCENT PREPARATION

PF

PM

- SEC F-PLN page As rqrd
 - a) If weather is OK, SEC F-PLN can be used for setting another possible approach and/or
 - b) RWY as a backup at destination airfield. If there is a last minute RWY change, it is then only necessary to activate SEC F-PLN not forgetting to set new MDA/DH and NAVAIDS.
- **CABIN CREW ADVISE**, PF announces: **"CABIN CREW PREPARE FOR APPROACH"**
- **APPROACH BRIEFING**; It is recommended to use FMS pages and ND as a guide for descent and approach briefing. Main points to be covered are :
 - a) Navaid - ILS, VOR selection procedures and crossing altitudes.
 - b) F-PLN page - STAR, APPR, TRANS, MISSED APPROACH
 - c) APPR page - Landing Config, speeds, MDA,
 - d) FUEL PRED - Fuel needed for diversion, holding, fuel available.
 - e) Runway conditions, lighting and dimensions, actual landing distance.
 - f) Ground Spoiler, reverse operation, autobrake selection and remotely tuned navaid.
 - g) Terminal area topography, transition level and minimum safe altitudes to ensure a proper terrain awareness.
 - h) Weather at destination.
 - i) Go Around:
 - o Standard call / task sharing,
 - o Diversion decision.
- **NOTE:** If AP disengaged, it is recommended to descend at Mach 0.8 or below to avoid alpha-trim activation.(0.78 is fleet policy).

DESCENT

PF

PM

DESCENT MONITOR
FMA CK / ANNOUNCE ANY CHANGES
PF CDU PROG

FMA MONITOR ANY CHANGES
PF CDU F-PLN

- Descent will be initiated automatically in profile mode at the Top of Descent (T/D) point, provided a lower altitude has been preset on the FCU.
- The FMS will hold the CRZ FL until the top of descent.
- 30 seconds before reaching the T/D, P.DES will flash to indicate that the descent will be initiated without any further pilot action.
- If ATC requires an early descent, The descent may be initiated immediately (before reaching T/D) by pressing IMM DES prompt on CDU. In this case the descent will be performed at 1 000 ft/min (this value can be changed on CDU) until the pre computed path is reached.
- If the descent is delayed by ATC, When passing the T/D the aircraft stays at the CRZ ALT as no lower altitude has been selected on the FCU. DECEL prompt is displayed in line 1R. **Pressing this prompt reduces speed to green dot (highest of PFD and FMS values if they are different).** As soon as clearance is given by ATC, select the cleared ALT on the FCU, select desired descent speed on the FCU, then pull the FCU speed/Mach knob. The descent will now be initiated in LVL!CH mode.
- DESCENT CLEARANCE Obtain
- WHEN DESCENT CLEARANCE IS GIVEN BY ATC:
 - a) FCU ALT KNOB TURN to select cleared altitude
 - b) FCU ALT KNOB Pull
 - c) P.THR / P.DES armed Check FMA
 - d) IMM DES prompt is displayed on FMS CDU.
- ANTI ICE As rqrd
 - a) **During descent ENG ANTI ICE should be ON when icing conditions are met, or when moderate to severe precipitation is encountered.**
- IGNITION As rqrd
 - a) **Ignition should be selected to CONT RELIGHT prior to ENG ANTI ICE selection.**
- PF CDU PROG
 - a) PF CDU should preferably be set to PROG page in order to see the VDEV and DIST TO DEST information. Other pages such as F-PLN may be selected as necessary.
- With AP engaged PF should make any required F-PLN revisions.
- PM CDU F-PLN

DESCENT

PF

PM

- Descent monitoring can be achieved as follows:
 - a) When flying in NAV mode, PROFILE mode is most probably used. P.DES indicated on the FMA. The aircraft descends along the descent flight path, VDEV is provided on PROG page, and may be thus monitored. All constraints of the FPLN will be taken into account by the FMS.
 - b) When flying in HDG (TRK) mode out of the lateral F.PLN, PROFILE DESCENT should not be used.
 - c) PROG page displays flight plan distance to DESTINATION and can display BRG/DIST to DESTINATION if selected. The comparison of this data is useful to monitor the descent.
 - d) The level "----" symbol, on ND may be used to monitor the descent as well.
 - e) The predictions on CDU assume a return to the FPLN path.
 - f) From time to time, during stabilized descent, select FPA on PFD and check that remaining distance to destination is approximately the FL change required divided by FPA in degrees.

$$FPA (^{\circ}) = \Delta FL / DIST (NM)$$
- When each new descent clearance is given by ATC
 - a) Cleared level Set on FCU
 - b) FMA indication Check/Announce
 - c) Cleared ALT on PFD Check
- Check MSA and ensure that the aircraft does not descend below MSA. Monitor progress in the descent, particularly when approaching the cleared altitude.
- When 1000 Ft above the cleared altitude.
 - a) Announce "one thousand to go"
- Monitor reduction in V/S as the cleared altitude is reached. In V/S or LVL/CH mode, monitor ALT* engagement, then ALT at the cleared altitude.

F.L. 200 MIN :

SEAT BELT.....ON
TERR ON ND.....AS RQRD

DESCENT C/L :

TERR ON ND.....AS RQRD

- At level off:
- FMA indication Check/Announce
- **NOTE:** In PROFILE mode, the descent is performed at idle thrust, maintaining the required descent path by pitch control. Thus the aircraft speed will vary with the pitch and is not directly controlled. However, if the speed varies by more than 20 kt from the FMS target speed (most probably due to lack of sufficient, or accurate descent wind inputs) the following will occur:
 - a) If the Speed is 20 kt above the target, or reaches VMAX-2 Kt, "MORE DRAG" flashes on the PFD. The SPD BRK should be extended. The message will flash until the speed is within 5 kt of the target.
 - b) If the speed is 20 kt below the target or 10 kt below Green Dot, and the SPD BRK is extended, the message "LESS DRAG" is displayed on the CDU. The SPD BRK should be retracted. If the SPD BRK is not extended, the aircraft reverts to P.SPD mode, power is increased and the target speed re-captured.

DESCENT

PF

PM

- **DESCENT ADJUSTMENT:**
 - If an increased rate of descent is desired :
 - a) Increase descent speed (by use of LVL/CH and selected speed) if comfort and ATC permit. This is economically better (Time / Fuel).
 - b) Maintain high speed as long as possible (SPD LIM may be cleared, subject to ATC clearance).
 - c) If aircraft is high with high speed, it is more efficient to keep high speed until level off and then decelerate rather than to mix descent and deceleration.
 - d) If the aircraft goes below the desired profile, use V/S mode to adjust rate of descent.
 - e) In descent speedbrake may be used to increase the rate of descent. Half speedbrake extension may be used to maintain the required rate of descent when engine anti ice is used.
 - f) In Profile descent (P.DES) if the aircraft is on or below the flight path and ATC requires an increased rate of descent, do not use speedbrake on its own since rate of descent is dictated by planned flight path. In this case select MAX DES mode on TACT page with speed brake.
 - **NOTE :**
 - 1) Use of full speedbrake above 0.78 M is uncomfortable for passengers.
 - 2) With ANTI-ICE ON engine thrust is increased which will reduce the descent path angle at idle. This can be compensated for by an increase of descent speed, or by extending half speedbrake.
 - 3) Do not use speedbrake in configuration 30/40. In configuration 15/20, high buffet may be expected if more than half speedbrake is selected. When speedbrake is extended, the V/s increases on PFD, which may hamper airspeed reduction. When landing in 15/20, the speedbrake must be retracted at the latest before reaching approach stabilization height in short final.
 - 4) Do not change speedbrake position during configuration changes.
 - **RADAR TILT :**
 - a) Every 10 000 ft during descent down to around 15 000 ft, adjust the tilt angle slightly upwards to eliminate excessive ground clutter on the upper part of the ND.
 - b) From 15 000 ft, adjust the tilt angle by one degree upwards per 5 000 ft descent, in order to keep the ND relatively free of ground clutter.
 - Consider switching TERR pushbutton to ON ND ,depending on external conditions (mountainous area, low ceiling, low visibility). If the use of radar is required, consider selecting radar display on PF side, and TERR ON ND on PM side only.

F.L. 150 EXT LTS ON

AT TRANSITION LEVEL :

ALTIMETERSET

TCAS THRT

APPROACH C/L :

NAV ACCURACYCK

ALTIMETER SET

CABIN CREWADVISE

- **CABIN CREW ADVISE, PM announces: "CABIN CREW LANDING POSITION"**
- Set QNH on all altimeters when approaching the transition level and when cleared for an altitude.
- Crosscheck baro setting and altitude readings.
- NOTE: When operating in low OAT , altitude corections as defined in 2.08.10 page 4 should be considered.

STANDARD APPROACH

INITIAL APPROACH

PF

PM

- The following approach procedure assumes AP engaged and A/THR engaged, which is the recommended procedure.
- Set ONH on all altimeters when approaching the transition altitude and when cleared for an altitude. Crosscheck baro settings and altitude readings.
- When reaching Terminal Area, (approximately 50 NM from the destination) crosscheck navigation accuracy.

IGNCONT RELIGHT / AS RQRD

POSITIONINGMONITOR
NAV ACCURACYCK

FCU / FMS SET (IF AP OFF)
POSITIONING MONITOR

CABIN REPORTOBTAIN (CM1)

- Select IGNITION rotary sel to CONT RELIGHT if required. It is recommended to select CONT RELIGHT if the landing runway is covered with standing water, in heavy rain, or if severe turbulence is expected in the approach or Go around area.
- Check aircraft positioning for a smooth transition to the approach. Rule of thumb : 9000 ft at 250 Kt IS 30 NM to touchdown.
- NAV/COM FREQ Check/Set
 - Set VHF as required
 - Set ILS frequency and course
 - Set VOR/NAV/ILS switch to ILS as required
- APPROACH SPEED
 - If ATC requires a particular speed to be flown then use selected speed. When ATC speed constraint no longer applies, return to profile mode.
- Monitor navigation accuracy and be prepared to change approach strategy, particularly when IRS ONLY NAVIGATION is displayed.
 - If navigation accuracy is greater than 1 NM ND must be used in ROSE or ARC mode by PF.
 - NAV mode should not be used, use HDG SEL.
 - Switch off the enhanced functions of the EGPWS via the TERR MODE pushbutton switch located on the Captain's switching panel.
- RADAR TILT: Increase tilt as required to keep the ND clear.
- ND Set/Check
- PF ND MODEMAP for ILS approach otherwise depending on the FMS accuracy
- PM ND MODE MAP may be kept
- ND RANGE Select 15 NM
- DH Check
- Both CM1 and CM2 confirm DH setting on EFIS control box in accordance with company policy.
 - For CAT 1 precision approaches, or non-precision approaches, set the CAT 1 DH or MDA using the altimeter amber index and set a DH =- 5 on the EFIS control panel to cancel the "MINIMUM" auto call-out.

STANDARD APPROACH

FINAL APPROACH

PF

PM

- The following approach procedure assumes AP engaged and A/THR engaged, which is the recommended procedure.
- Conducting a stabilized approach is recommended. The objective is to be stabilized on the final descent path at VAPP in the landing configuration, at 1500 ft AAL.
- To be stabilized, all of the following conditions must be achieved prior to, or upon reaching this stabilization height :
 - The aircraft is on the correct lateral and vertical flight path,
 - The aircraft is in the desired landing configuration,
 - The power is stabilized and the aircraft is trimmed to maintain VAPP on the desired glide path,
 - No excessive flight parameter deviation.
- The advantages are :
 - Lower fuel consumption
 - Lower noise levels
 - Time saving
 - Flexibility and ability to vary speed to suit ATC.
- If the aircraft is not stabilized on the approach path in landing configuration at 1000 ft AAL in IMC, or at 500ft AAL in VMC, or as restricted by Operator policy/regulations, a go-around must be initiated unless the crew estimates that only small corrections are necessary to rectify minor deviations from stabilized conditions due, amongst others, to external perturbations.

FCU*....."GREEN DOT" SPEED

ILS.....SELECT & IDENTIFY
HDG SEL.....AS RQRD
SLATS EXTEND.....ORDER

WHEN SFPI SHOWS SLATS 15 :

ILS.....SELECT & IDENTIFY
ANNOUNCE "SPEED CHECKED"
SLAT 15..... SELECT
ANNOUNCE..... "SLATS EXTENDED"

FCU*....."S" SPEED

- Slats 15 should be extended not later than 3NM prior to the FAF (Final Approach Fix)
- Check deceleration towards S speed. The aircraft should be established on the glide slope with Slats 15 at "S" Speed above 2 500 ft AAL.
- In the event that A/C speed is significantly higher than "S" on the G/S, or the A/C does not decelerate on the G/S, extend the L/G in order to slow the A/C down.

EGPWS LDG SLATS / FLAPS SW.....AS RQRD
NAV ACCURACY.....CK

- If landing is planned to be performed in S 15°/F 20° landing configuration select GPWS LANDING FLAPS switch to 20.
- Landing with Flaps 20 is recommended in windshear conditions or if approach climb requirements cannot be met with Flaps 40.

WHEN CLEARED FOR FINAL APP :

LAND..... PRESS
2ND AP.....ENGAGE
FMACK / ANNOUNCE
LOC CAPTURE.....MONITOR
"LOC* - RWY HEADING".....ANNOUNCE

ILS COURSE.....CK

*PF if AP on, PM if AP off.

STANDARD APPROACH

FINAL APPROACH

PF

PM

- LAND pb on FCU Press
a) This enables LOC and G/S capture.
- **Note:** Arm the LAND mode only when ATC gives clearance for the ILS approach (clearance for both LOC and G/S capture). If only LOC capture required, use V/L mode until clearance for full ILS received.
- At LOC capture, NAV or HDG is disengaged automatically. Check correct ILS course set.

RWY HDG*.....SET

G/S CAPTUREMONITOR
"G/S* - MISSED APPROACH ALT...FT SET" ...ANNOUNCE

ANNOUNCE "CHECKED"

GO AROUND ALT*.....SET

- If above the glideslope
a) V/S mode Select
b) FCU altitude Set above aircraft altitude
- **NOTE:** TOGA thrust limit is automatically selected on TRP when slats are extended and at least one AP/FD engaged. Then, at light weights, another thrust limit (CLB or CRZ) can be manually selected to avoid excessive pitch attitude and rate of climb.
- Between 2500 ft and 2000 ft AGL
a) RAD ALT CHECK
b) Announce "RAD ALT ALIVE"
- **NOTE:** Crew should now keep RA in scan to landing.

AT 2500 FT AAL MIN :

FLAPS 15ORDER

WHEN SFPI SHOWS FLAPS 15 :

ANNOUNCE "SPEED CHECKED"

FLAPS 15 SELECT

ANNOUNCE "FLAPS 15"

FCU*....."F" SPEED

FLAPS 20ORDER

WHEN SFPI SHOWS FLAPS 20 :

L/G DOWNORDER

FLAPS 40ORDER

WHEN SFPI SHOWS FLAPS 40 :

ANNOUNCE "SPEED CHECKED"

FLAPS 20 SELECT

ANNOUNCE "FLAPS 20"

L/GDOWN

GND SPLRARM

AUTO BRKAS RQRD

CK / ANNOUNCE "GEAR DOWN"

ANNOUNCE "SPEED CHECKED"

FLAPS 40 SELECT

ANNOUNCE "FLAPS 40"

FCU*....."VAPP"

*PF if AP on, PM if AP off.

STANDARD APPROACH

FINAL APPROACH

PF

PM

LANDING C/L

Announce any FMA modifications

AT 1000 ft AAL :

AT 700 ft AAL :

AT 400 ft :

"LAND".....CK/ ANNOUNCE

AT DA + 100 ft :

AT DA :

ANNOUNCE..... "CONTINUE" or "GO AROUND FLAPS"

AUTO CALL OUT.....MONITOR

Check any FMA modifications

FLT PARAMETERS.....CK

ANNOUNCE "1000"

ANNOUNCE..... "COURSE SET"

ANNOUNCE "CHECKED"

ANNOUNCE..... "100 ABOVE"

MONITOR/ ANNOUNCE "MINIMUMS"

- Check "3 green" on landing gear main indication panel.
 - BRAKE YELLOW HYDRAULIC PRESSURE CHECK
 - a) If residual pressure is indicated: Depress several times the brake pedals until release of residual pressure.
 - b) If residual pressure remains: Select preferably the AUTOBRAKE system or apply brakes at touchdown.
 - c) A slight brake deflection (3°) will supersede any previous yellow pressure.
 - **Note** : If antiskid is inoperative maintain a symmetrical braking as soon as aircraft touchdown.
 - AUTO BRKAs reqrd
 - a) The use of autobrake is recommended. Select the appropriate pushbutton according to the runway length and conditions and check related ON light illuminated.
 - b) The use of MAX is not recommended at landing.
 - c) When landing on short or contaminated runway or when operating in low visibility weather conditions use MED mode.
 - d) On a normal runway length LO mode is recommended.
 - e) It is recommended to use max reverse thrust until 80 kt. Then idle reverse and brakes, as necessary, according to the remaining distance.
 - **Note** : If, on very long runways, the pilot anticipates that braking will not be needed, use of the autobrake is unnecessary.
 - ECAM wheel pageCheck, Check 8 brake release indications.
 - ALL AVAILABLE LIGHTS ON: Selecting lights on even in daylight will minimize bird strike hazard.
 - FLIGHT PARAMETERS Check:
 - o PF announces any FMA modification
 - o PM calls out:
 - a) Pitch attitude becomes lower than 00 or greater than 10° nose up.
 - b) Bank angle becomes greater than 7°.
 - c) V/S greater than 1 000 ft/min.
 - d) Airspeed deviations of more than + 10 kt or - 5kt.
 - e) Any significant changes in ground speed that might indicate windshear.
- Note: Maximum allowable lateral deviation at final segment is 1 DOT and maximum allowable vertical deviation at final segment is 1/2 DOT up to D/A.
No deviation is allowed afterwards.

STANDARD APPROACH

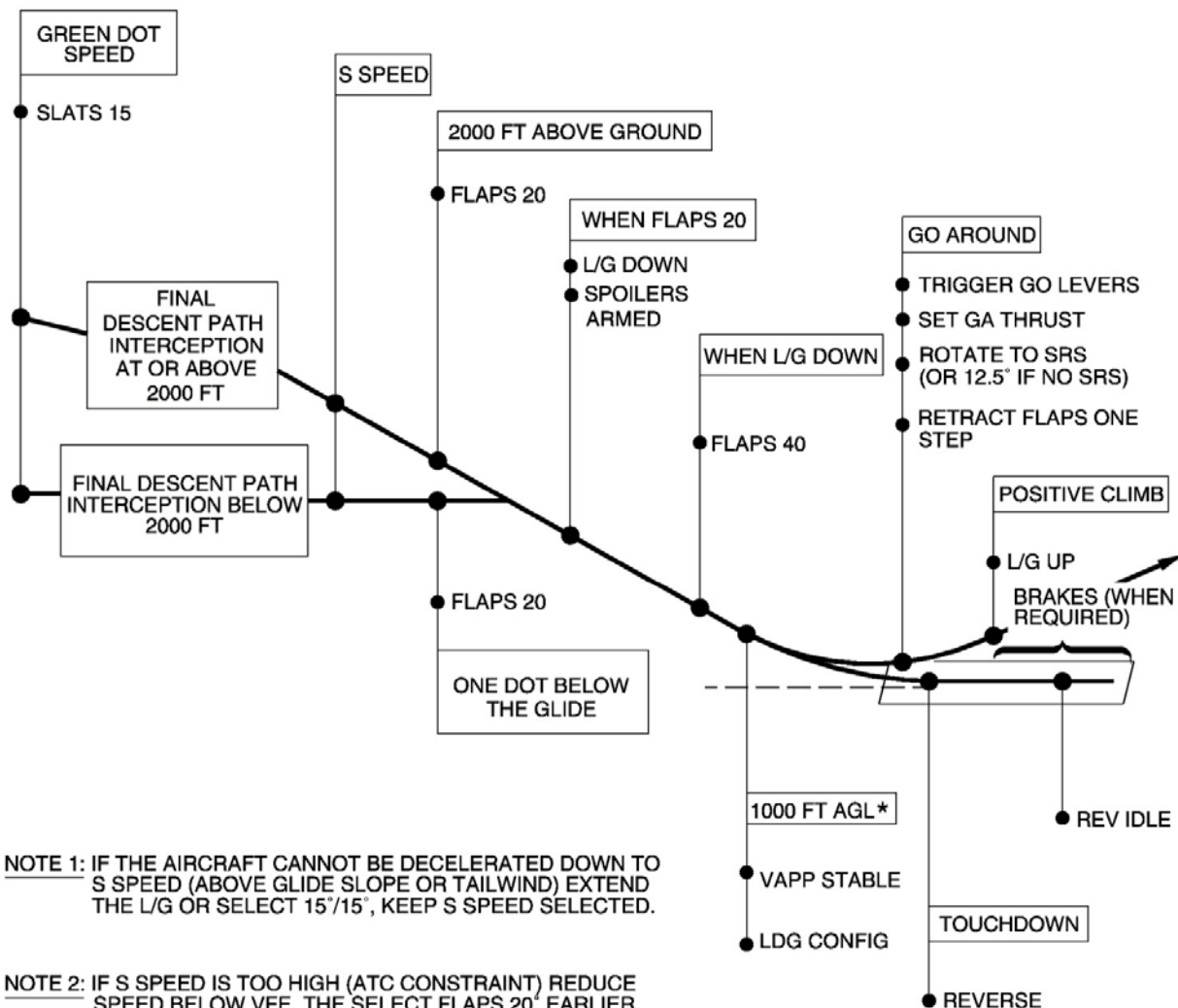
FINAL APPROACH

PF

PM

- When the PM calls flight parameter exceedance, the suitable **PF** response is:
 - a) Acknowledge the **PM** call out, for proper crew coordination purposes.
 - b) Take immediate corrective action to control the exceeded parameter back into the defined stabilized conditions.
 - c) Assess whether stabilized conditions will be recovered early enough prior to landing, otherwise initiate a Go Around.
- **WHEN VISUAL**; Do not "duck under" the G/S. Maintain a stabilized flight path down to the flare.
- **NOTE:** Close to the ground, avoid important down corrections. Give priority to attitude and sink rate.
- **AUTO CALL-OUT** Monitor, or announce appropriate heights as per company policy.
- **NOTE:** At 50 ft, aircraft one dot below GIS is 7 ft below G/S.

STANDARD APPROACH



CPCS_02_03_19_004_A003_00

VISUAL APPROACH

PF

PM

- **INTRODUCTION:** This procedure provides general guidelines to perform a visual approach. In every case, if a visual approach has been published for the landing runway, it must be applied.
- **Note :** Keep in mind the possible risk of optical illusions due to night vision.
- **OBJECTIVE:** The objective is to approach using visual references on a nominal 3 degree glideslope and to be stabilized at 500ft AAL. To be stabilized, all the following conditions must be achieved before or when reaching this stabilization height :
 - a) The aircraft is on the correct lateral and vertical flight path,
 - b) The aircraft is in the desired landing configuration,
 - c) The power is stabilized and the aircraft is trimmed to maintain VAPP on the desired glide path,
 - d) No excessive flight parameter deviation.
- If the aircraft is not stabilized on the approach path in landing configuration at 500 ft AAL, or as restricted by Operator policy/regulations, a go-around must be initiated unless the crew estimates that only small corrections are necessary to rectify minor deviations from stabilized conditions due, amongst others, to external perturbations.
- **METHOD FOR VISUAL APPROACH:**
 - a) The autopilot should be disconnected.
 - b) The use of the FPV is recommended.
- **INITIAL / INTERMEDIATE APPROACH:**
 - o The flight plan selected on the CDU should include the selection of the LANDING RWY.
 - o The down wind leg might be also part of the F PLN. This may be a useful indication of the aircraft positioning in the circuit, on the ND. However visual references must be used.
 - o Therefore, at beginning of down wind leg:
 - a) Select FPV ON
 - b) The use of A/THR is at pilot's discretion.
 - c) Apply the VISUAL APPROACH flight pattern.
 - d) Down wind leg extension 45 seconds(± wind correction).
 - e) Turn into base leg Bank 30° max.
- **FINAL APPROACH:**
 - a) The speed trend arrow and FPV are useful for achieving timely and correct thrust (if in manual thrust) and approach path corrections. Avoid descent through the correct approach path with idle thrust. (Late recognition of this situation without prompt thrust increase may lead to considerable speed decay and altitude loss).
 - b) Avoid any tendency to "duck under" in the latter stages of the approach.
 - c) Avoid destabilization of the approach in the last 100ft to give the best chance of achieving a good touch down at the desired position.
- **NOTE :** Close to the ground, avoid important down corrections. Give priority to attitude and sink rate.

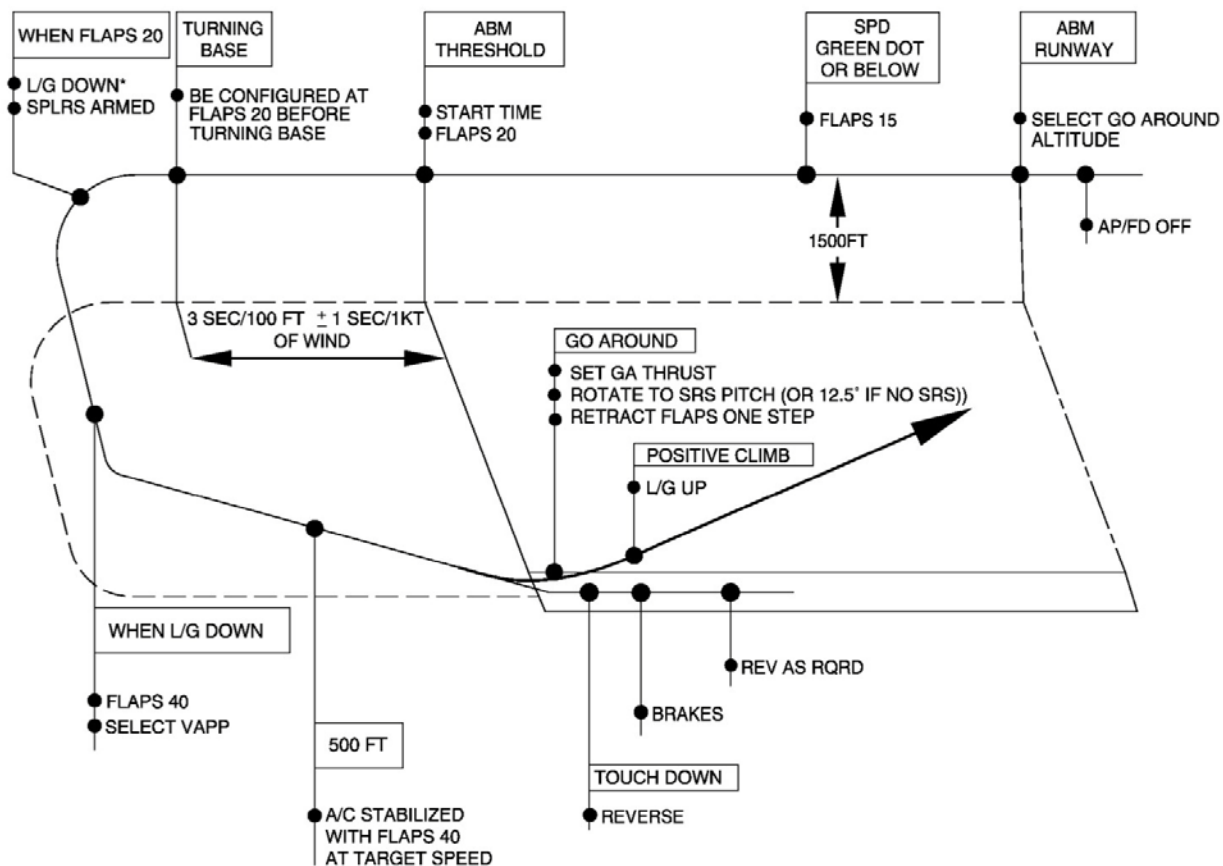
VISUAL APPROACH

PF

PM

VISUAL APPROACH (1 or 2 engines) PATTERN

VISUAL APPROACH



CFCs_02_03_20_006_A001_00

* CONSIDER SINGLE ENGINE OPERATIONS

NON PRECISION APPROACH

PF

PM

- **INTRODUCTION:** This procedure contains recommendations that are specific to Non Precision Approaches. The standard approach procedure steps have not been repeated. They must be completed before the Final Approach Fix.
- This procedure provides general guidelines which may be adapted according to the airline policy and/or the actual flight conditions.
- **APPLICABILITY:** This procedure applies to all non-ILS approaches, e.g. NDB, VOR, VOR-DME, LOC ONLY, LOC-DME. For LOC BACK COURSE refer to Procedures and Techniques chapter Use of AFS.
- **APPROACH GUIDANCE:** Non Precision Approaches can be performed using two different AP/FD guidance :
 - o FMS guidance :
 - a) NAV mode down to the MDA or until LOC interception (lateral).
 - b) PROFILE mode until FAF then V/S mode down to the MDA (vertical).
 - o Selected guidance :
 - a) HDG SEL mode down to the MDA (lateral) or until LOC interception.
 - b) V/S mode after leaving the FAF down to the MDA (vertical).
- When the approach is stored in the NAV database and the NAV ACCURACY check is positive, the FMS guidance is recommended. Otherwise, the selected guidance must be used.
- In both cases, the recommended flight reference display for a Non Precision Approach is the FPV/FPR.
- **APPROACH SPEED TECHNIQUE:**
 - a) The standard speed technique is a stabilized approach using AP engaged in CMD mode and A/THR engaged in SPD mode.

INITIAL APPROACH

NAVIGATION ACCURACY CK
REFERENCE NAVAIDS TUNED / CHECKED
MDA (MDH) CK / SET

- Check navigation accuracy using NAVAIDS raw data to determine approach strategy.
- For VOR approach, tune frequency and course on VOR control panel and check VOR is also tuned on CDU.
- Set MDA+50 ft, (MDH+50 ft) using the altimeter amber index and set a DH -5 on the EFIS control panel to cancel the "MINIMUM" auto call out.

If accuracy check is positive

NDMAP

ND MAP

- Use NAV mode down to the MDA or until LOC interception (lateral).
- Use PROFILE mode until FAF then V/S mode down to the MDA (vertical).

VOR/NAV/ILS switchNAV

VOR/NAV/ILS switchNAV

If accuracy check is negative

NDROSE or ARC

NDMAP or ROSE or ARC

- Use HDG SEL mode down to the MDA or until LOC Interception (lateral).
- Use V/S mode after leaving the FAF down to the MDA (vertical).

VOR/NAV/ILS switchVOR or ILS
FPA/CRS targetSET

VOR/NAV/ILS switchVOR or ILS
FPA/CRS targetSET

APPROACH C/L COMPLETE

NON PRECISION APPROACH

PF

PM

INTERMEDIATE APPROACH

- The STANDARD APPROACH steps to prepare the aircraft for landing (L/G down, spoilers armed, Flaps 40, speed VAPP) should be performed before reaching the FAF. If the aircraft is not stabilized on the approach path in landing configuration at 1000 ft AAL in IMC, or at 500 ft AAL in VMC, or as restricted by the operator policy / regulations, A go-around must be initiated unless the crew estimates that only small corrections are necessary to rectify minor deviations from the stabilized condition due, amongst others, to external perturbations.

FINAL APPROACH

FD/FPV switch.....SELECT FPV

FD/FPV switch.....SELECT FPV

- Select FPV just before FAF at the latest. The FPR is displayed only if the VOR/NAV/ILS switch is in VOR or ILS position.

AT FAF

V/S SELECT

If ND is in MAP mode

VOR/NAV/ILS switchILS
GA ALTITUDESET

VOR/NAV/ILS switchILS

- ILS must be selected for NDB and LOC approaches VOR must be selected for all other non precision approaches.
- Set GA altitude, when below the go around altitude.

During Final Approach

POSITION & FLIGHT PATH.....CK / ADJUST

- POSITION and FLIGHT PATH CHECK/ADJUST
 - Monitor reference NAVAID raw data.
 - Monitor altitude in relation with the published descent profile and the distance to the runway.
 - Adjust HDG SEL and V/S accordingly.
 - If FMS NAV is not satisfactory, revert to HDG/SEL.

LANDING C/L COMPLETE

FLIGHT PARAMETERSCK

FLIGHT PARAMETERS.....CK

Announce any FMA modifications

- PF announces any FMA modification.
- PM calls out if:
 - Speed becomes lower than VAPP- 5 kt or greater than speed target +10 kt.
 - Pitch attitude becomes lower than - 2.5° or greater than 10° nose up.
 - Bank angle becomes greater than 7°.
 - Descent rate becomes greater than 1000 ft/min.
 - Any significant changes in ground speed that might indicate windshear.

Note: Maximum allowable lateral deviation is 2 DOTs during final segment of approach and maximum allowed vertical deviation is +200/-0 ft on final stage of approach for non precision approaches and Altitude deviation at DA or MDA: 0 ft.

NON PRECISION APPROACH

PF

PM

FINAL APPROACH

AT 1000 ft AAL :
AT 700 ft AAL :
AT MDA (MDH) + 100 ft

ANNOUNCE "1000"
ANNOUNCE "COURSE SET"
ANNOUNCE "100 ABOVE"

- When the PM calls flight parameter exceedance, the suitable **PF** response is :
 - Acknowledge the **PM** call out, for proper crew coordination purposes.
 - Take immediate corrective action to control the exceeded parameter back into the defined stabilized conditions.
 - Assess whether stabilized conditions will be recovered early enough prior to landing, otherwise initiate a Go-Around.

Approaching MDA (MDH)

When visual references are acquired & confirmed by both PF/PM

ANNOUNCE "CONTINUE"
AP OFF
FPR OFF

FPR OFF

- Continue as visual approach with the standard call outs.
- Note:** Close to the ground, avoid important down corrections. Give priority to attitude and sink rate.

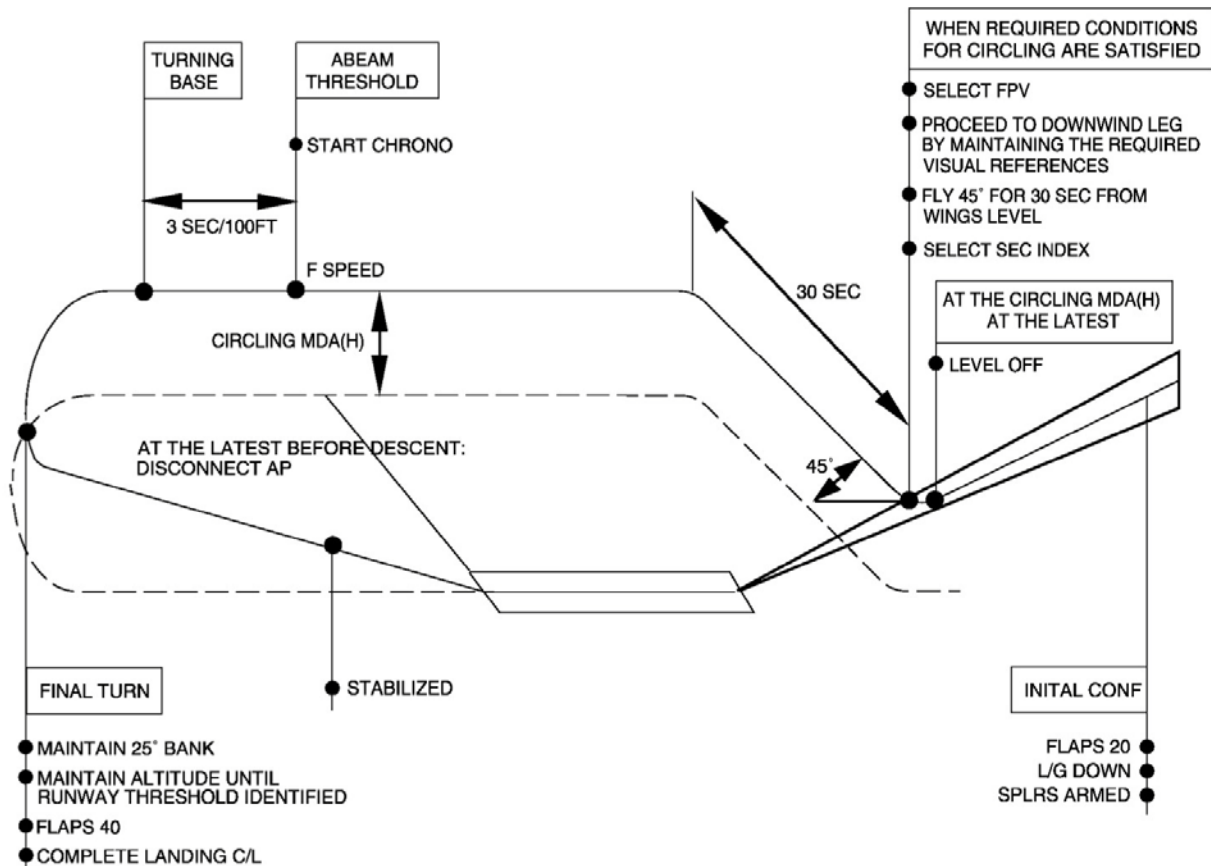
If no visual references are acquired

ANNOUNCE "GO AROUND/FLAPS"
AUTO CALL OUT MONITOR

- Note:** If ground references are not visible when the aircraft reaches MDA, an immediate Go-Around must be initiated.

**NON PRECISION APPROACH
LOW VISIBILITY CIRCLING APPROACH**

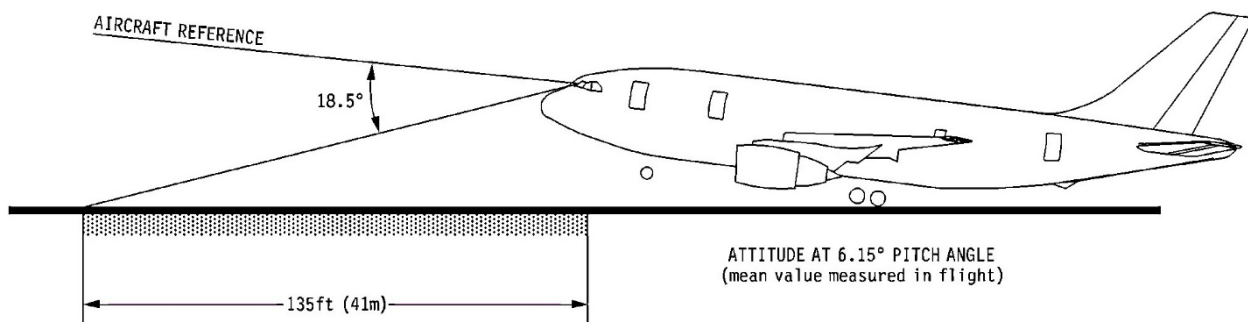
LOW VISIBILITY CIRCLING APPROACH PATTERN



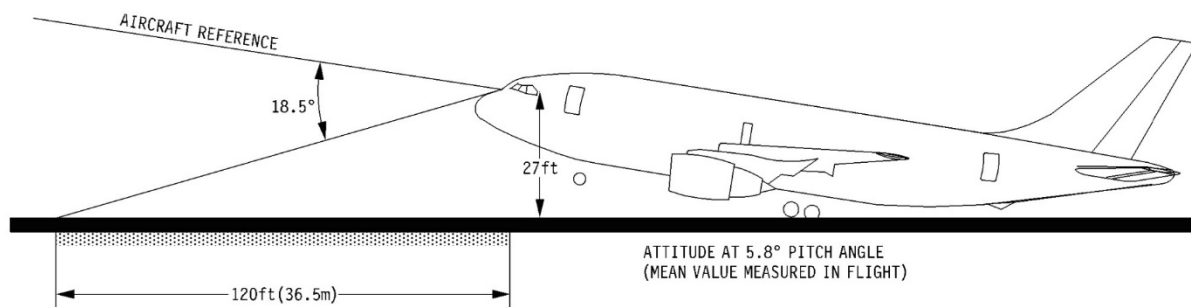
CFC5_02_03_20_005_A001_00

LANDING

PILOT EYE ORIENTATION WITH GROUND



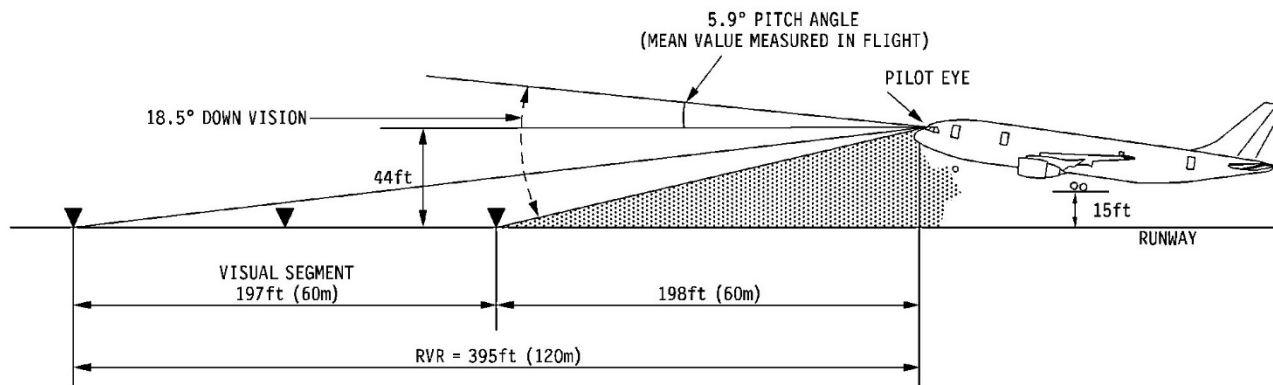
A300-600R



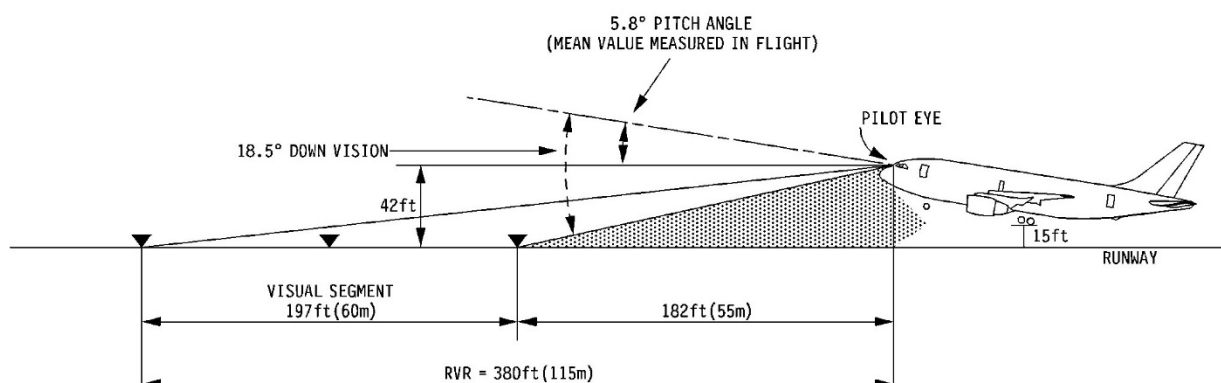
A310

LANDING

VISUAL GROUND SEGMENTS LANDING APPROACH H=15ft



A300-600R

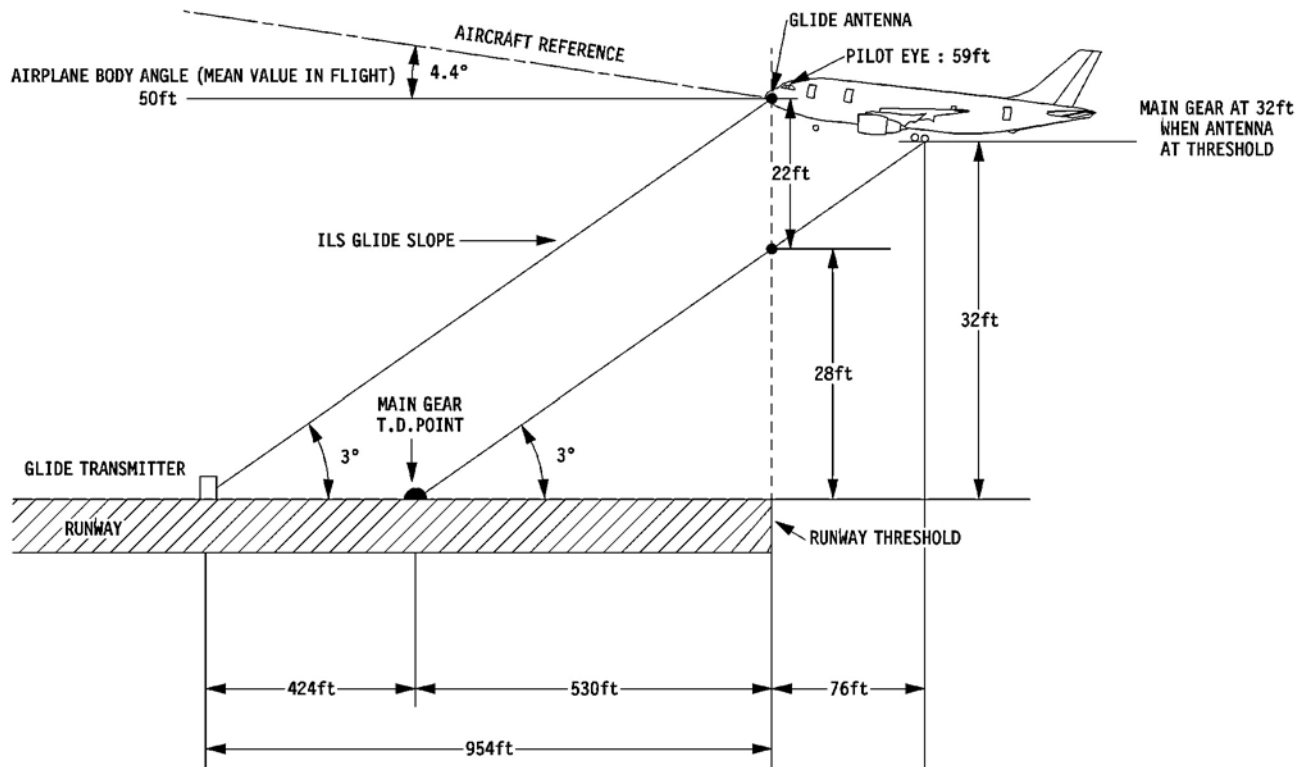


A310

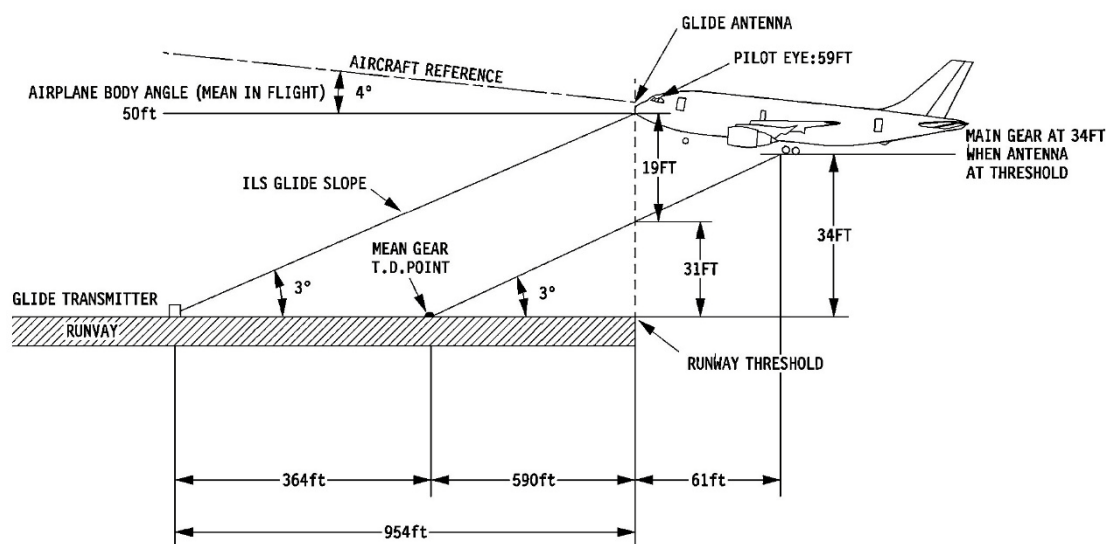
LANDING

ESTIMATED ILS APPROACH

Airplane following straight line extension of glide at threshold



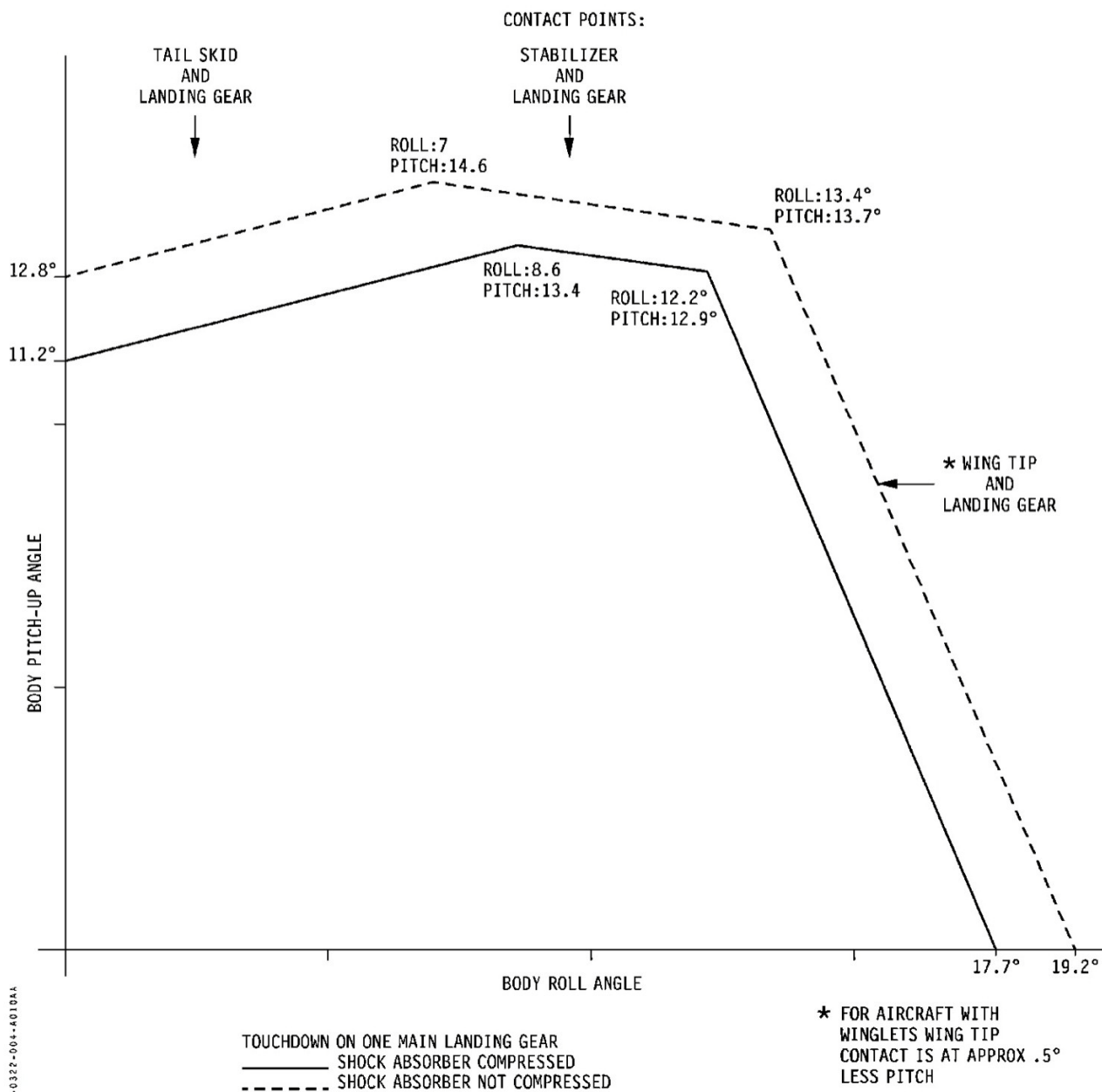
A300-600R



A310

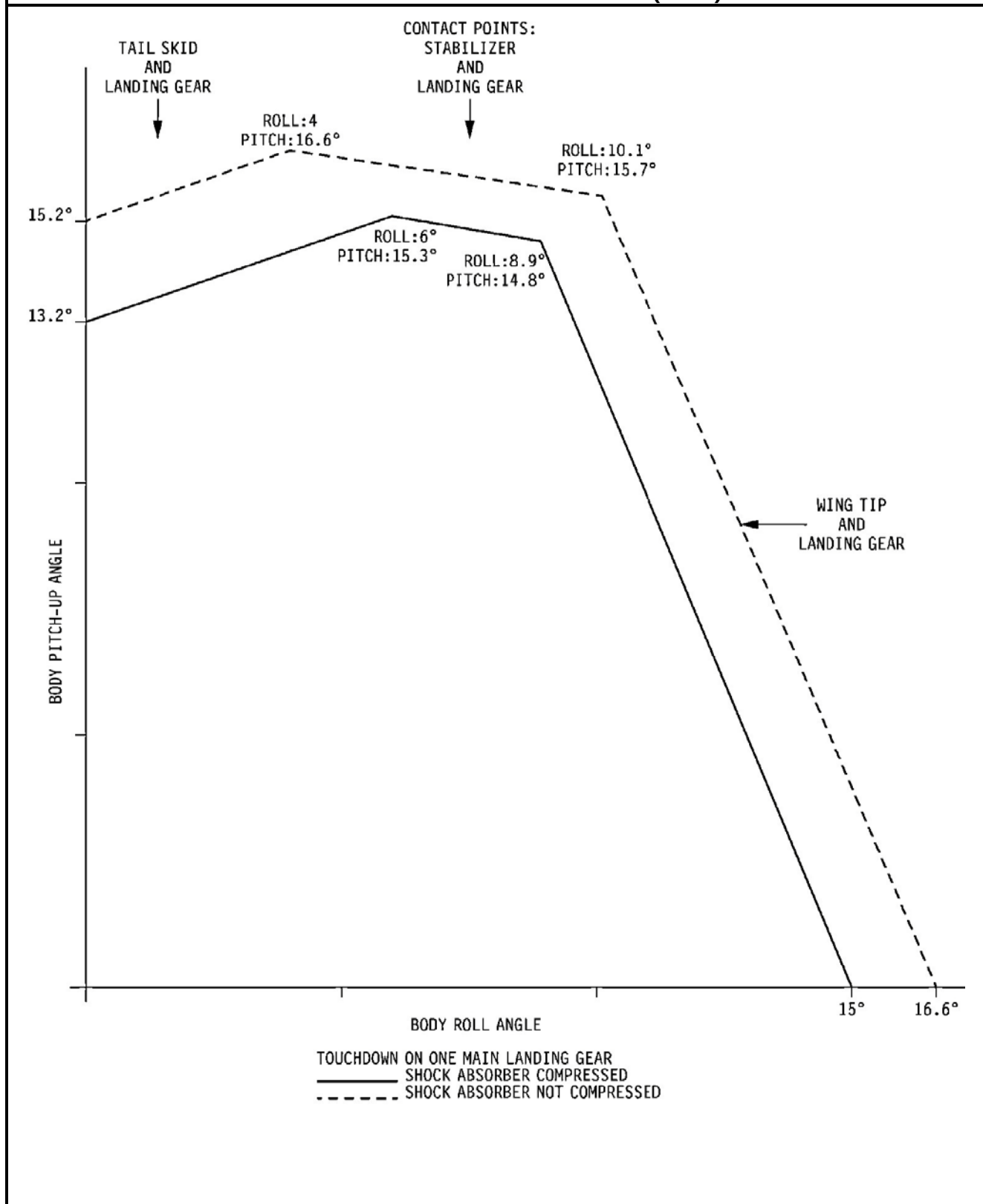
LANDING

GROUND CLEARANCE DIAGRAM (A300-600R)



LANDING

GROUND CLEARANCE DIAGRAM (A310)



LANDING

PF

PM

- From stabilized approach conditions, the flare height is about 30 ft.

FLARE.....PERFORM
THROTTLES.....MONITOR IDLE

- PM** should monitor the attitude and call out :
 - a) **PITCH PITCH**, when pitch reaches 9°.
 - b) **BANK BANK** when bank reaches 7°.
- NOTE:** Do not flare too high or with too high pitch attitude. Tail strike will occur at +11° pitch. The pilot's objective, with respect to vertical navigation, is to maintain a constant flight path angle down to the runway threshold, using the vertical deviation indication of an ILS. However, when approaching flare height, the pilot's primary objective will shift from vertical flight path control to safe pitch attitude and vertical speed to start the flare in good conditions.
- Monitor throttles reduction to idle at 30 ft.
- If A/THR is not engaged, retard throttles to idle.
- If A/THR is engaged, it will automatically disconnect at touchdown provided that both throttles are at idle.
- NOTE:** If one or both thrust levers remain above the IDLE position, ground spoilers extension is inhibited.

AT TOUCHDOWN

REVERSE PULL

NI, EGT, IAS.....MONITOR
GND SPLRS EXT.....CK / ANNOUNCE
REVERSERS.....CK / ANNOUNCE
DECELERATION.....CK / ANNOUNCE

- Immediately after touch down of main landing gear, pull reverse levers to the idle mechanical stop.
- After reverse thrust is initiated, a full-stop landing must be performed.
- NOTE:** Maximum efficiency of the reverse is obtained at high speed.
- NOTE:** Maintain a slight backward pressure on reverse levers in order to identify the release of the idle mechanical stop. Do not move release levers towards stow position while reversers are in transit; such action may cause system damage.
- Check ground spoilers extension after touch down on ECAM system display.
- If no ground spoilers are extended :
 - a) Verify and confirm that both thrust levers are at idle position.
 - b) Set both thrust levers to REV MAX, and simultaneously fully press the brake pedals.
 - c) Extend speed brake.
- Check reverse deployment is as expected (REV green).
- The deceleration is felt by the flight crew, and confirmed by the speed trend on the PFD.
- NOTE:** If ground spoilers are not armed they extend at reverser thrust selection.

REVERSE THRUST.....AS RQRD
DIRECTIONAL CONTROL.....RUDDER PEDALS
BRAKES.....AS RQRD

- NOTE:** If the use of max reverse is restricted due to noise consideration, maintain reverse idle until taxi speed is reached.

LANDING

PF

PM

- **NOTE:** If directional control problems are encountered, reduce thrust to reverse idle until directional control is satisfactory.
- **CAUTION:** Except in an emergency, the use of high levels of reverse thrust at low airspeed should be avoided since the distortion of the air flow caused by gases re-entering the compressor can cause engine stalls which may result in excessive EGT.
- **CAUTION:** Do not use NWS Control handle before taxi speed is reached.
- Apply brakes as required or monitor autobrake. When autobrake is used the **PF** should be ready to takeover and apply manual braking as required.
- **NOTE:** If no ground spoilers are extended, the autobrake is not activated.
- **CAUTION:** If brakes are found inoperative, switch immediately BRK-A/SKID to ALTN/OFF and modulate brake pressure as required at or below 1000 PSI. Brake pedals should be released when the A/SKID is switched OFF. Otherwise the pedal braking orders will be taken into account and the aircraft will react strongly.

AT 80 Kts OR IAS FLUCTUATION

REVERSE IDLE

ANNOUNCE "80 Kts"

AT TAXI SPEED

REVERSESTOW

- Stow the reverse when taxi speed is reached and before leaving the runway.
- **CAUTION:** Do not recycle if reverse fails to stow. Engine shut down is recommended.
- **CAUTION:** On taxiways, the use of reverse, even restricted to idle thrust, may have the following effects:
 - a) Fine sand and debris may be ingested which might be detrimental to both the engine and airframe systems.
 - b) On snow covered areas, snow will be recirculated into the air inlet, which may result in engine flame out or roll back.
- **CAUTION:** Except in an emergency, reverse thrust should not be used to control aircraft speed while taxiing.
- **NOTE:** The flight crew must select **reverse thrust** immediately after main landing gear touchdown.
The flight crew must immediately select REV MAX, if any of the following occurs at any time during the landing:
 - An emergency
 - The deceleration is not as expected, or
 - A failure affects the landing performance, or
 - A long flare or a long touchdown
 - An unexpected tailwind.
- A small pitch up may occur during thrust reversers deployment before nose landing gear touchdown. However, the flight crew can easily control this pitch up. As soon as the flight crew selects reverse thrust, the flight crew must perform a full-stop landing.

GO AROUND

PF

PM

- When "GO AROUND-FLAPS" is called, **PM** retracts FLAPS one step.
- NOTE:** Triggering one GO LEVER is sufficient to initiate GO AROUND.
- Follow through on THROTTLE LEVERS if ATS is armed or manually set GO AROUND thrust if ATS is not armed.

ANNOUNCE "GO AROUND - FLAPS"
GO LEVERS TRIGGER
FMA - CK / ANNOUNCE "THR - GO AROUND"
THROTTLE LEVERS FOLLOW THROUGH TOGA
ROTATION PERFORM
ORDER "GEAR UP"

FLAPS RETRACT ONE STEP
ANNOUNCE "CHECKED"
ANNOUNCE "POSITIVE CLIMB"
L/G SELECT UP
L/G LEVER NEUTRAL
ANNOUNCE "GEAR UP / NEUTRAL"

- CAUTION:** PF should be ready to override or disconnect the autothrottle function in case of thrust asymmetry and to counteract aircraft yaw.
- Rotate the aircraft at a rate of typically 3 degrees per seconds.
- Initiate rotation towards 18° of pitch with all engines operative (approximately 12.5° if one engine is out) to get a positive rate of climb then follow SRS Flight Director pitch bars orders if available (not exceeding 18 degrees) or maintain pitch.
- As thrust increases, be prepared to counteract the thrust pitch-up moment.
- Trim the aircraft nose down as required. The pitch attitude should not be allowed to develop beyond 25 degrees, as such a pitch attitude would result in a significant speed loss.
- An immediate and firm elevator nose down command (together with a nose down pitch trim order) would allow to recover the target pitch attitude.
-
- NOTE:** FMS CDU automatically switches to GO AROUND page.
- Monitor:
 - The airspeed and speed trend,
 - The pitch attitude and bank angle
 - The thrust increase (GA thrust on the TRP and on N1/EPR indicators).
- A call out must be made in case of excessive deviation :
 - Speed dropping below VLS-5KT
 - Speed trend negative
 - Pitch attitude in excess of 20° up or lower than 10° up
 - Bank angle in excess of 15 degrees (30 degrees if the missed approach procedure requires a turn)
 - Significant thrust loss.

NAV or HDG* SELECT

AT THRUST REDUCTION ALTITUDE

TRP* CL or CR

- THROTTLES Check symmetrical retard movement
- NOTE:** In case of asymmetrical throttle retard movement, A/THR should be disconnected and thrust manually set.

*PF if AP on, PM if AP off.

STANDARD OPERATING PROCEDURE

A300-600R/A310

P 61

GO AROUND

PF

PM

AT ACCELERATION ALTITUDE

SPD*.....250 KT / G.A. SPD

LVL/CH*.....SELECT

MISSED APPROACH PROCEDUREFOLLOW

FLAPSRETRACT ON SCHEDULE

*PF if AP on, PM if AP off.

AFTER LANDING	
PF	PM
EXT LTS.....AS RQRD GND SPLRS.....DISARM	ICE PROTECTIONAS RQRD IGN..... OFF ΔP CK XPNDR/TCAS.....MODE "S" WX RADAR.....OFF/TEST PITCH TRIM1° NOSE UP SLATS/FLAPS.....RETRACT BRAKE TEMPCK BRAKE FANSAS RQRD
- The following steps should be done once the aircraft has cleared the runway. - Select the STROBE lights AUTO when leaving the runway. - Select LDG lights OFF. - Select TAXI light to TAXI. - It is recommended to retract FLAPS step by step to minimize the possibility of jamming (mainly on bumpy taxiways). - If the approach was made in icing conditions, or, if the runway was contaminated with slush or snow, do not retract the slats/flaps until after engine shutdown once the ground crew has confirmed them clear of obstructing ice. - On the ground, hot weather conditions may cause overheating to be detected around the bleed ducts in the wings, resulting in " BLEED LEAK " warnings. Such warnings may be avoided during transit by keeping the slats / flaps in Configuration 15/0 when OAT is above 30°C. - Check brake temperature on ECAM wheel page for discrepancy or high temperature. - A report for maintenance action must be made when : a) 1. Temperature difference between two brakes on the same gear is greater than 150°C AND b) Temperature of one brake is below 60°C (loss of braking capability on this wheel) or temperature of one brake is above 600°C (brake binding on this wheel). OR c) 2. Difference between average temperature of LH and RH main landing gear brakes is greater or equal to 200°C (no significant crosswind, or differential braking). Brake fans selection should be delayed for a minimum of about 5 minutes or just before stopping at the gate (whichever occurs first), to allow for thermal equalization and stabilization and thus avoid oxidation of brake surface hot spots. However, when turn around times are short, or brake temperatures are likely to exceed 500°C, use the brake fans disregarding possible oxidation phenomenon. NOTE: The arc displayed on the ECAM WHEEL page, suggesting the selection of brake fans at taxi out, should be ignored for taxi in. - The "BRAKES TEMP HI" ECAM warning after landing should not lead to the immediate selection of brake fans as long as 500°C is not likely to be reached (wait 5 minutes or gate arrival).	
AFTER LANDING C/L (SILENCE)	
Empty space for C/L (SILENCE) section	

PARKING

PF

PM

- Turn **NOSE TAXI** light OFF before turning towards ground marshaller approaching stand.
- The **ACCU PRESS** indication must be in the green band. In case of low accumulator pressure, chocks are required before engines shutdown.

PARK BRK ACCU PRESS CK
PARK BRK..... ON
GPU SIDE ENG FUEL LEVER OFF
EXT PWR ESTABLISH
2nd ENG FUEL LEVER..... OFF

ATC XPNDRSTBY

- With **Parking Brake ON**, if the aircraft starts to move due to a parking brake failure, immediately release the Parking Brake and restore braking using the pedals.
- **NOTE:** If BRK HOT light is illuminated, the flight crew should ensure chocks are in place as soon as possible and then release the PARKING BRAKE.
- **NOTE:** Selection of APU BLEED before engine shut down avoids cycling of the packs. However delay as late as possible to avoid engine exhaust fumes entering the air cond.
- **NOTE:** If APU is not available, set EXT PWR to ON before shutting down engines.
- **NOTE:** For thermal stabilization before shutdown, the engines should be operated at idle or required taxi thrust until gate arrival. Refer to Single Engine Taxi procedure, as required.
- **NOTE:** If N2 does not decrease upon FUEL LEVER selection to OFF (HP VALVE light illuminated) select hydraulic pumps OFF and pull fire handle. Engine will shut down after 70 to 90 sec.

AFTER BOTH ENGINES SPOOL DOWN SIGNIFICANTLY

"ENGs SHUTDOWN" ANNOUNCE
BEACON OFF
EXT LTS.....AS RQRD
SEAT BELT.....OFF
FUEL PUMPS.....OFF
WINDOW HEAT.....AS RQRD
PROBE HEAT.....OFF
GROUND CONTACT.....ESTABLISHED
PARK BRK.....OFF / AS RQRD

ENGs SHUTDOWN.....CK & CONFIRM

SLIDES DISARMED.....CK

- Set all lights as required. Switch off BEACON once all engines have spooled down. **N2 below 20%**
- Check on ECAM DOOR page. Warn cabin crew if a slide is not disarmed.
- Check chocks in place and release parking brake to improve cooling.
- It is recommended not to use the parking brake for prolonged periods when brake temperatures are above 200°C to avoid hydraulic fluid degradation.
- **NOTE:** Do not select RUD TRAVEL pushbutton switch to OFF.
- Set all fuel pumps to OFF except L INNER TANK Pump 2 if fuel remains in INNER TK and APU is used.
- TRIM TK MODE Pushbutton Check AUTO
- **FUEL QUANTITY:**
 - a) Check that the sum of the Fuel On Board and the Fuel Used is consistent with the Fuel On Board at departure. If an abnormal discrepancy is found, maintenance action is due.
 - b) Record the Fuel On Board for crosscheck prior to the next flight.
- **NOTE:** Window Heat can be selected ON during the turn-around time to avoid ice build-up on the frontal windshield and the lateral windows.

PARKING	
PF	PM
ELAPSED TIME.....STOP CRT'S AS RQRD PARKING C/L	ELAPSED TIMESTOP IRS DRIFT CK IRS MSU 1,2,3, OFF / AS RQRD BRK TEMPCK BRK FANSAS RQRD CRT'SAS RQRD
- Record IRS 1, 2, 3 position, position error, Ground speed or drift rate as required for maintenance monitoring. NOTE: GS is set to zero 2 minutes after aircraft has stopped moving. Thus GS must be recorded before this time. NOTE: Dim the CRTs to increase their life, do not switch OFF, If aircraft is not flying again within 2 hours. Switch EFIS CRTs to OFF.	
SECURING THE AIRCRAFT	
PF	PM
SECURING THE AIRCRAFT C/L	IRS MSU 1,2,3,.....OFF CREW OXYOFF EXT LTSAS RQRD EMER EXIT LTDISARM BATTERIES..... OFF
- After switching the IRS OFF, wait at least 10 seconds before switching off the electrical supply to ensure that the IRS memorize the last position data. If APU was used: a) Set master switch to OFF after the passengers have disembarked. b) Set L INR TK PUMP 2 to OFF. CAUTION: If APU is running when leaving the aircraft do not switch OFF the BATTERIES.	

STANDARD CALLS

COMMUNICATIONS AND STANDARD TERMS

- Standard phraseology is essential to ensure effective crew communication. The phraseology should be concise and exact. The following chapter lists calls that should be used as standard. They supplement the call outs identified in the SOP.
- These Airbus standard calls are designed to promote situational awareness, particularly each time one crewmember adjusts or changes information and/or equipment on the flight deck (e.g. FMA change, FMS alteration, change in speed or Mach, tuning nav aids, flight path modification, system selection). These Airbus standard calls are also designed to ensure that the flight crew understands the systems and their use in line operation.

CHECKLIST CALLS

- "CHECK" : A command for the other pilot to check an item.
- "CHECKED" : A response that an item has been checked.
- "CROSS-CHECKED" : A call verifying information from both pilot stations.
- "HOLD CHECKLIST AT" : If a checklist needs to be interrupted.
- "RESUME CHECKLIST AT" : Announce for continuation.
- "CHECKLIST COMPLETED" : Announce at the completion of a checklist.

ACTIONS COMMANDED BY PF

- The following commands do not necessarily initiate a change in a guidance mode. The intent is to ensure clear, consistent, standard communication between crew members.
- All actions made on the FCU and CDU must be verified on the PFD and ND as follows
 - a) First ensure that the correct FCU knob is used, then verify indications on the PFD/ND.
 - b) Mode changes should be confirmed by calling the color when appropriate, - e.g. BLUE.
- **SET:** The command "SET" means to use a knob to set a value, but not to change a mode. SET is accomplished by only rotating the appropriate selection knob :
 - a) "SET GO AROUND ALTITUDE"
 - a) "SET ONH"
 - b) "SET ALTITUDE"
- **ENGAGE:** The command "ENGAGE" means to PUSH the specified FCU button or set the specified lever or switch to "ON" that causes an immediate FMA change:
 - a) "ENGAGE NAV"
 - b) "ENGAGE PROFILE"
 - c) "ENGAGE ALT HOLD"
- **ARM:** The command "ARM" means to arm a system, such as by pushing the specified FCU button.
 - a) "ARM LAND"
 - b) "ARM LOC"
- **PULL:** The command "PULL" means to use a FCU knob to engage a mode.
 - a) "ALTITUDE 8000 PULL"
 - b) "VERTICAL SPEED SET -1000FT/MN AND PULL"
- **PUSH:** The command "PUSH" means to use a FCU knob to synchronize or preset a value.
 - a) "HEADING PUSH"
 - b) "SPEED PUSH PRESET 250"
- **NOTE:** Whenever there is no requirement for PULL/PUSH, and if a new setting is required, simply command "SET" the specific selector knob followed by the required value.
- (existing mode is lateral mode heading of 140 degrees and a new heading of 100 degrees is required; command should be "SET HEADING 100".

STANDARD CALLS

- **FMA:** Unless listed otherwise (e.g. CAT II & III task sharing), all FMA changes will be normally called by the PF and checked by the PM:
 - a) All armed modes are announced by calling their associated color e.g.: "G/S BLUE", "LOC BLUE".
 - b) All active modes are announced without calling the color (green, white) e.g. : "NAV", "ALT".
- **NOTE:** If the FMA change is not announced by the PF, the PM should announce it.
- **ALTITUDE:** The PM calls "ONE THOUSAND TO GO" when passing 1000 feet before the cleared altitude or FL, and the PF calls "CHECKED".

SLATS/FLAPS CONFIGURATION

- After selection of the Slats/Flaps lever position, the PM replies when the correct selection is shown on SFPI.
- **EXTENSION:** PM Checks the speed below VFE next and decelerating (approach).

SLATS/FLAPS CONFIGURATION	PF	PM
0/0 to 15/0	"SLATS EXTEND"	"SPEED CHECKED " (When SFPI shows 15/00) "SLATS EXTENDED"
15/0 to 15/15	"FLAPS 15"	"SPEED CHECKED " (When SFPI shows 15/15) "FLAPS 15"
15/15 to 15/20	"FLAPS 20"	"SPEED CHECKED " (When SFPI shows 15/20) "FLAPS 20"
15/20 to 30/40	"FLAPS 40"	"SPEEDCHECKED " (When SFPI shows 30/40) "FLAPS 40"

- **RETRACTION:** PM Checks the speed above the "S" or "F" speed and accelerating (takeoff).

SLATS/FLAPS CONFIGURATION	PF	PM
SLATS/FLAPS 15/20 passing "F" speed 15/20 to 15/0	"FLAPS 0"	"SPEED CHECKED SELECTING FLAPS 0" (When SFPI shows 15/00) "FLAPS 0"
SLATS 15 passing "S" speed 15/0 to 0/0	"SLATS RETRACT"	"SPEED CHECKED SELECTING SLATS 0" (When SFPI shows 00/00) "SLATS RETRACTED"

- **GEAR CONFIGURATION:**

EVENT	PF	PM
To retract the landing gear	"GEAR UP"	"GEAR UP/NEUTRAL" The PM selects the gear lever position and replies when lights are extinguished and gear lever set to Neutral.
To extend the landing gear	"GEAR DOWN"	"GEAR DOWN" The PM selects the gear lever position and replies after checking 3 green lights on the landing gear main indication panel.

STANDARD CALLS

FLIGHT PARAMETERS

- PM will make calls for the following conditions during final approach. Attitude callouts are also to be made through to landing.

CONDITIONS	PM
When airspeed becomes less than VAPP- 5 or more than speed target +10	"SPEED"
When V/S is greater than -1000 ft/min.	"SINK RATE"
When bank angle becomes greater than 7°	"BANK"
When pitch attitude becomes lower than 00° or higher than +10° nose up. For non precision approach, when pitch attitude becomes lower than -2.5° or higher than +10° nose up.	"PITCH"
When excessive LOC or GLIDE deviation occurs : 1/4 dot LOC 1 dot GS	"LOC" or "GLIDE SLOPE"
When greater than 0,5 dot (VOR) or 5 degrees (ADF).	"COURSE"
At altitude checks points.	"-----FT HIGH (LOW)"

PF/PM DUTIES TRANSFER

- To transfer control, flight crewmembers must use the following calls:
 - To give control: The pilot calls "YOU HAVE CONTROL". The other pilot accepts this transfer by calling "I HAVE CONTROL", before assuming PF duties.
 - To take control: The pilot calls "I HAVE CONTROL". The other pilot accepts this transfer by calling "YOU HAVE CONTROL", before assuming PM duties.

ABNORMAL AND EMERGENCY CALLS

- ECAM Procedure:**
 - "ECAM ACTION" is commanded by PF when required.
 - "CLEAR (title of the system)" is asked by the PM for confirmation by the PF, that all actions have been taken/reviewed on the present ECAM WARNING/CAUTION or SYSTEM PAGE.
e.g. : "CLEAR HYDRAULIC".
 - "CLEAR (title of the system)" is the command by the PF that the action and review is confirmed. For status page, the call "REMOVE STATUS" is used.
 - "ECAM ACTIONS COMPLETED" is the announcement by the PM that all APPLICABLE ACTIONS have been completed and system pages have been reviewed.
 - Should the PF requires an action from the PM during ECAM procedures, the order "STOP ECAM" is used. When ready to resume the ECAM the order/statement "CONTINUE ECAM" is used.

MEMORY ITEMS

- The following commands allow the crew to be aware of a situation that requires or will require actions to be done by memory. Crew will be prepared to properly react in terms of crew coordination, task sharing and communication.

CALLS	CONDITIONS
"PULL UP TOGA"	When GPWS warning require an avoidance maneuver
"WINDSHEAR TOGA"	When windshear conditions required to select maximum takeoff power
"UNRELIABLE SPEED"	When an unreliable speed indication is detected
"TCAS, I have control"	"TRAFFIC" warning is triggered by TCAS
"EMERGENCY DESCENT"	When EMERGENCY DESCENT has to be initiated
"LOSS OF BRAKING"	In case of loss of braking at landing or during taxi

STANDARD CALLS

SUMMARY FOR EACH PHASE

EXTERNAL POWER DISCONNECTION

EVENT	PF	GND MECH
Initial ground contact	"GROUND (from) COCKPIT"	"COCKPIT (from) GROUND"
External disconnection	"REMOVE EXTERNAL"	"EXTERNAL REMOVED"

BEFORE ENGINE START

EVENT	PF	PM
Before startup clearance received	"BEFORE START CHECKLIST"	"DOWN TO THE LINE"
After startup clearance received	"BELOW THE LINE"	"BEFORE START C/L COMPLETED"

PUSH BACK

EVENT	PF	GND Mech
When ready for push back and push back clearance received from ATC	"GROUND (from) COCKPIT" "READY FOR PHSH"	"COCKPIT (from) GROUND" "RELEASE PARKING BRAKE"
Start of push	"BRAKES RELEASED CLEAR TO PUSH"	
When push back completed	"PARKING BRAKES SET"	"SET PARKING BRAKES"
When ready to disconnect (Engine parameters are stabilized)	"CLEAR TO DISCONNECT"	"DISCONNECTING" "HAND SIGNAL ON THE LEFT/RIGHT"

ENGINE START

EVENT	PF	GND MECH
When start up clearance obtained	"CLEAR TO START, STARTING ENGINE 2/1"	"CLEAR TO START ENGINE 2/1"
When fan starts to rotate		"FAN ROTATION"

AFTER ENGINE START

EVENT	PF	PM
All engines started and stabilized and GND is disconnected	"AFTER START C/L"	"AFTER START C/L COMPLETED"
Flight controls check in following sequence	"FLIGHT CONTROL CHECK"	
Elevators		"FULL UP" "FULL DOWN" "NEUTRAL"
Ailerons/Spoilers		"FULL LEFT" "FULL RIGHT" "NEUTRAL"
Rudder	"RUDDER"	"FULL LEFT" "FULL RIGHT" "NEUTRAL"

- **NOTE:** PM should follow pedal movement with feet.

TAXI/BEFORE TAKE OFF

When taxi clearance obtained	"CLEAR LEFT(RIGHT) SIDE"	"CLEAR RIGHT(LEFT) SIDE"
Checking Brakes	"BRAKE CHECK"	"PRESSURE ZERO"
During TAXI	"BEFORE TAKE OFF CHECKLIST"	"DOWN TO THE LINE"
Lining up on the runway	"BELOW THE LINE"	"BEFORE TAKE OFF C/L COMPLETED"

STANDARD CALLS

TAKE OFF

EVENT	PF	PM
When triggering go levers	"TAKE OFF" Announce FMA	"CHECKED"
Before passing 80 kts		"THRUST SET"
At 100 kts	"CHECKED"	"ONE HUNDRED"
AT V1		"V1"
AT VR		"ROTATE"
When positively clear from the ground (V/S positive RAD ALT increase)	"GEAR UP"	"POSITIVE CLIMB"
If AP is required	"ENGAGE AP 1(2)" Followed by confirmation on FMA	"AP 1(2) ENGAGED"
When L/G retracted		"GEAR UP/NEUTRAL"
When "F" speed and accelerating	"FLAPS 0"	"SPEED CKECKED SELECTING FLAPS 0"
When "S" speed and accelerating	"SLATS RETRACT"	"SPEED CHECKED SELECTING SLATS 0"
After TAKE OFF/CLIMB C/L	"AFTER TAKE OFF C/L DOWN TO THE LINE"	"AFTER TAKE OFF C/L DOWN TO THE LINE COMPLETED"
At transition altitude	"AFTER TAKE OFF C/L BELOW THE LINE"	"AFTER TAKE OFF C/L COMPLETED"

REJECTED TAKE OFF

EVENT	CAPTAIN	F/O
RTO Decision	"STOP"	
GND SPLR Extension		"SPOILERS"
MAX reverse thrust selection		"REVERSE GREEN"
Deceleration		"DECEL"

- **NOTE:** If ground spoiler deployment is not as expected, call "NO GROUND SPOILER"
- **NOTE:** If reverse deployment is not as expected call " NO REVERSE ENG 1(2), or "NO REVERSE" as appropriate.
- **"DECEL"** call means that the deceleration is felt by the crew and confirmed by the speed trend on the PFD.
- If **AUTOBRAKE** is armed and no flow bar green light observed, call "NO AUTOBRAKE".

CLIMB, CRUISE, DESCENT PHASES

- During climb, cruise and descent phases communication should be as standard terms.

ALTIMETER SETTING CHANGES TO/FROM QNH/QFE-STD

EVENT	PF	PM
Barometric setting change and subsequent altimeter cross-check	"PULL STANDARD" "CHECKED"	"STANDARD CROSS-CHECKED PASSING FL-----NOW"
Barometric setting change and subsequent altimeter cross-check	"SET ONH/OFE" "CHECKED"	"ONH/OFE----- CROSS-CHECKED PASSING----- FT NOW"

STANDARD CALLS		
APPROACH AND LANDING		
EVENT	PF	PM
When cleared below transition level or as appropriate	"APPROACH C/L"	"APPROACH C/L COMPLETED"
Initial approach	"SET GREEN DOT SPEED" *	"GREEN DOT SPEED SET"
Beginning of radio altimeter indication	"CHECKED"	"RADIO ALTIMETER ALIVE"
NOTE: CREW SHOULD KEEP "RA" IN SCAN TO LANDING.		
At Green Dot speed or below VFE	"SLATS EXTEND" "SET "S" SPEED" *	"SPEED CHECKED" (When SFPI shows SLATS 15) "SLATS EXTENDED"
At "S" speed or below VFE	"FLAPS 15" "SET "F" SPEED" *	"SPEED CHECKED" (When SFPI shows FLAPS 15) "FLAPS 15"
At "F" speed or below VFE	"FLAPS 20"	"SPEED CHECKED" (When SFPI shows FLAPS 20) "FLAPS 20"
Interception Heading	"ARM LAND" *	"LAND ARMED"
Localizer Capture	"LOC STAR" "SET RUNWAY HEADING" *	"----° SET"
2 DOTS ALL ENGINE 1 DOT SINGLE ENGINE	"GEAR DOWN"	"GLIDE SLOPE ALIVE" (When L/G is down) "GEAR DOWN"
When gear is down	"FLAPS 40" "SET VAPP" *	"SPEED CHECKED" (When SFPI shows FLAPS 40) "FLAPS 40" "VAPP SET"
Glide Slope Capture	"GS STAR" "SET GO AROUND ALTITUDE" *	"-----FT SET"
When flaps 40	"LANDING CHECKLIST"	"LANDING CHECKLIST COMPLETED"
At 1000 feet AAL	"CHECKED"	"ONE THOUSAND"
At 700 feet AAL (For precision APP & non precision APP) Each pilot checks the ILS selected course on his ND		"COURSE SET"
At 400 feet AAL For precision APP	"LAND"	"CHECKED"
100 feet above MDA/DA (MDH/DH)	"CHECKED"	"ONE HUNDRED ABOVE"
MDA/DA (MDH/DH) Visual reference	"CONTINUE"	"MINIMUM"
MDA/DA (MDH/DH) NO visual reference	"GO AROUND FLAPS"	"MINIMUM NO RUNWAY"
After touchdown		"SPOILERS" "REVERSE GREEN"
When auto brake armed		"DECCEL"
At 80 knots or IAS fluctuations		"EIGHTY KNOTS"

*PF if AP on, PM if AP off.

STANDARD CALLS

GO AROUND

EVENT	PF	PM
GO AROUND decision	"GO AROUND FLAPS"	
Gear retraction	"GEAR UP"	"POSITIVE CLIMB" (When L/G is up) "GEAR UP/NEUTRAL"
Checklist	"AFTER TAKE OFF C/L"	"DOWN TO THE LINE"
At transition altitude	"BELOW THE LINE"	"AFTER TAKE OFF C/L COMPLETED"

AFTER LANDING

Checklist		"AFTER LANDING C/L SILENCE COMPLETED"
-----------	--	--

PARKING

Checklist	"PARKING C/L"	"PARKING C/L COMPLETED"
-----------	---------------	-------------------------

SECURING THE AIRCRAFT

Checklist	"SECURING THE AIRCRAFT C/L"	"SECURING THE AIRCRAFT C/L COMPLETED"
-----------	-----------------------------	--

GREEN OPERATING PROCEDURES

GENERAL

INTRODUCTION TO THE GREEN OPERATING PROCEDURES

This section describes the available fuel-reduction procedures during the different flight phases. Operators can evaluate these procedures to define their own fuel-reduction policy as part of general cost-reduction policy. Each operator is responsible for the decision of which costs to reduce, for example (non-exhaustive list):

- Maintenance costs
- Fuel consumption
- Operating costs
- Passengers comfort.

For example, Cost Index (CI) and Flexible Takeoff aim to reduce general costs not fuel consumption. The information provided in this chapter is not applicable to all operators to the same extent, and depends on their specific operations (geographical location, airport conditions local regulations, etc.).

The flight crew decides the procedures to apply, in accordance with the operator guidelines and current flight conditions.

PRELIMINARY COCKPIT PREPARATION

APU:

APU START DELAY
Use the ground electrical unit and ground pneumatic unit when available.

AIR CONDITIONING:

- If one PACK is sufficient to supply air during the preliminary cockpit preparation.
- One PACK VALVE -OFF
Change between the packs on each flight, in order to prevent excessive wear of one pack.

COCKPIT PREPARATION

FMS INITIALIZATION:

INIT PAGE:

Cost Index (CI)

Cost Index Check consistency with CFP
The (CI) is the ratio of time costs versus fuel costs. The (CI) does not aim to reduce fuel consumption.

Tropopause

Tropopause value AS RQRD
Check that tropopause value is correct, in order to ensure the accuracy of FMS predictions.

TAKEOFF PAGE:

Thrust Reduction Altitude (THR RED ALT) / Acceleration Altitude (ACCEL ALT).

- If conditions and regulations permit:

THR RED ALT AS LOW AS POSSIBLE
The lowest thrust reduction altitude that can be selected is 1500 ft AAL.

ACCEL ALT AS LOW AS POSSIBLE
The acceleration altitude must not be lower than the thrust reduction altitude.

In order to save fuel, the flight crew should reduce altitude for both thrust reduction and acceleration.

ATC CLEARANCE:

- If conditions and performance permit:

TAKEOFF RUNWAY - OPTIMIZE
At airport with reduced traffic, request takeoff from the runway which reduces the taxi and/or flight time.
Refer to FCOM 2.01.20 p3 Operating Performance Limitations.

BEFORE PUSHBACK or START

APU:

APU START DELAY
Delay APU start as long as possible.

PUSHBACK / START UP CLEARANCE:

ENGINES START DELAY
Ask for a start clearance as late as possible.

AFTER START

APU:

- If the APU is necessary during or after the taxi:

APU BLEED KEEP ON
When the engines are running, exhaust gas ingestion may occur during use of the APU bleed.

GREEN OPERATING PROCEDURES

AFTER START

AIR CONDITIONING:

- If one PACK is sufficient to supply air during taxi:

One PACK VALVE ----- -OFF

Change between the packs on each flight, in order to prevent excessive wear of one pack.

PACK FLOW SELECTION

Note : **ECON FLOW is only active in flight.**

ANTI-ICE:

The selection of anti-ice increases engine thrust and fuel consumption. Therefore, the flight crew should only select the anti-ice when necessary.

TAXI

THRUST ----- AS LOW AS POSSIBLE

Avoid excessive thrust application.

BEFORE TAKEOFF

AIR CONDITIONING:

PACK VALVE 1 + 2 ----- -OFF

Select packs to OFF, in order to reduce fuel consumption. During flexible takeoff, the selection of packs to OFF R reduces takeoff EGT and therefore maintenance costs.

TAKEOFF

AIR CONDITIONING:

- If packs have been set to OFF for takeoff:

- At thrust reduction altitude:

One PACK VALVE ----- ON

The selection of one pack to ON before the TRP CLIMB selection results in an EGT increase.

- When slats retracted:

Second PACK VALVE ----- ON

CLIMB

ECON SPEED:

PROFILE MODE ----- -USE

Unless restricted by the ATC, use PROFILE MODE during climb, in order to fly at the optimum ECON speed.

ACCELERATION BELOW 10 000 FT

By default, the FMS flight plan takes into account the 250 kts speed limitation below 10000 ft. If the ATC permits, remove this limitation, in order to accelerate and save fuel.

CRUISE

PACK FLOW SELECTION:

ECON FLOW ----- ON

The Procedures and Techniques - Air Conditioning chapter 2.02.02 provides the pack flow settings. When passenger numbers permit, ECON mode saves up to 0.8 % of fuel.

OPTIMUM FLIGHT LEVEL (OPT FL):

TARGET FL ----- OPT

During cruise, the OPT FL increases while the aircraft weight decreases. Monitor the OPT FL, and fly at the most appropriate flight level for optimum aircraft performance and fuel consumption.

WIND UPDATE:

WIND INFO ----- -CHECK

Accurate and up-to-date wind information enables the FMS to optimize:

- Fuel Predictions
- Determination of the OPT FL

Update wind information when the change is more than either:

- 30 ° in direction, or
- 30 kt in speed.

DESCENT SPEED MODE:

PROFILE MODE ----- -USE

Fly in PROFILE mode, in order to:

- Fly at ECON speed
- Optimize speed in accordance with the (C I) and current flight conditions.

The flight crew should not modify the (C I) in flight.

GREEN OPERATING PROCEDURES

DESCENT PREPARATION

LANDING PREPARATION:

When landing performance permits, the best combination to reduce both fuel costs and brake wear is:

CONF 15/20 + REV IDLE + AUTO BRAKE LO.

If the flight crew needs to reduce the landing distance, they should use the deceleration devices in the following order:

1. Consider CONF 30/40

2. Consider REV MAX

3. Consider Autobrake MED

The use of the Autobrake MED in combination with CONF 15/20 and REV IDLE increases brakes oxidation, which may be severe.

When the flight crew rides the brakes (and overrides the Autobrake) at landing, severe oxidation may occur.

DESCENT

DESCENT SPEED MODE:

PROFILE MODE - - - - - USE

Fly in PROFILE mode (P.SPD/P.DES), in order to:

- Fly at ECON speed
- Optimize speed in accordance with the (CI) and weather conditions

ANTI-ICE:

The selection of anti-ice increases engine thrust and fuel consumption. Therefore, the flight crew should only select the anti-ice when necessary.

HOLDING

FLAPS CONFIGURATION:

FLAPS - - - - - -AS RQRD

Clean configuration is the optimum configuration for a holding circuit.

When required (holding pattern or speed limitation), the flight crew may consider the selection of CONF 15/0.

OPTIMUM SPEED:

- In CONF CLEAN:

OPTIMUM SPEED - - - - - Green Dot

In clean configuration, fly at Green Dot speed, in order to optimize the lift-to-drag ratio.

APPROACH

DECELERATION:

- When conditions and ATC permit:

APPROACH - - - - - DECELERATED

A decelerated approach reduces fuel consumption.

When the approach type does not enable to fly a decelerated approach, the flight crew should fly an early stabilized approach

AIRCRAFT CONFIGURATION:

FLAPS - - - - - -AS RQRD

When landing performance permits, the selection of CONF 15/20 reduces both the approach time and fuel consumption.

AFTER LANDING

APU:

- If both engines are running:

APU START - - - - - DELAY

Delay the start of the APU as long as possible after landing.

GREEN OPERATING PROCEDURES

AFTER LANDING

AIR CONDITIONING:

- If one PACK is sufficient to supply air after the landing:

One PACK VALVE ----- OFF

Change between the packs on each flight, in order to prevent excessive wear of one pack.